US012415316B2

(12) United States Patent
Rossing

(10) Patent No.: US 12,415,316 B2

(45) Date of Patent: Sep. 16, 2025

(54) **FILAMENT FEEDING ASSEMBLY FOR FUSED FILAMENT FABRICATION SYSTEM**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 149 days.

(21) Appl. No.: **18/553,846**

(22) PCT Filed: **Apr. 26, 2022**

(86) PCT No.: **PCT/NL2022/050221**

§ 371 (c)(1),
(2) Date: **Oct. 4, 2023**

(87) PCT Pub. No.: **WO2022/231421**

PCT Pub. Date: **Nov. 3, 2022**

(65) **Prior Publication Data**

US 2024/0181706 A1 Jun. 6, 2024

(30) **Foreign Application Priority Data**

Apr. 28, 2021 (NL) 2028080

(51) **Int. Cl.**
B29C 64/336 (2017.01)
B29C 64/118 (2017.01)
(Continued)

(52) **U.S. Cl.**
CPC ***B29C 64/336*** (2017.08); ***B29C 64/118*** (2017.08); ***B29C 64/209*** (2017.08);
(Continued)

(58) **Field of Classification Search**
None
See application file for complete search history.

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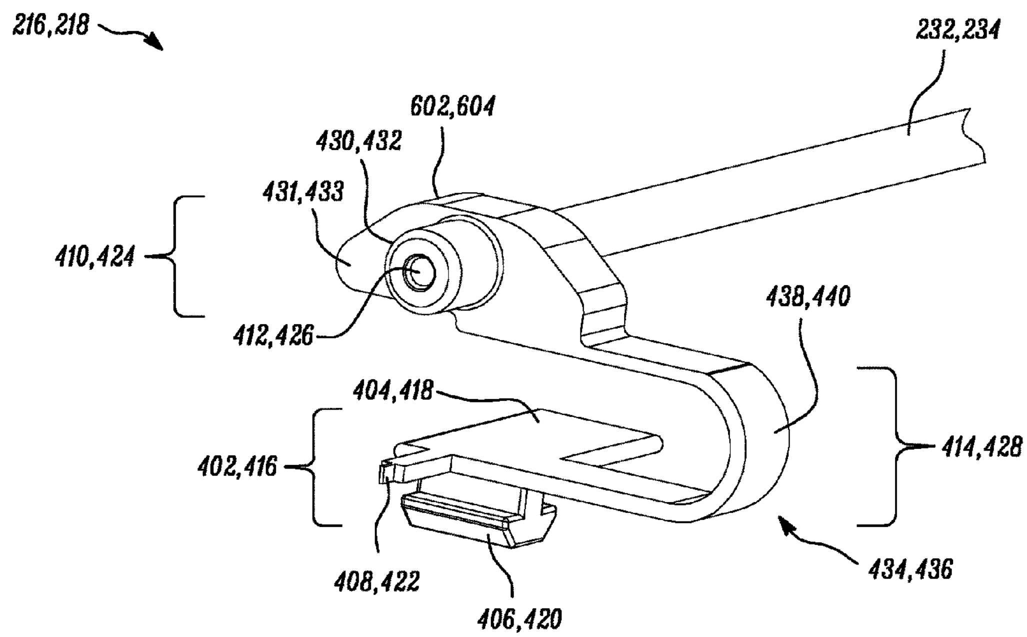
(57) **ABSTRACT**

Provided herein is a filament feeding assembly with a plurality of filament feeders arranged next to each other in a row. The filament feeding assembly has a first guide rail and a second guide rail, both arranged parallel to a straight line. The filament feeding assembly has first and second connectors movably coupled to the first and second guide rails, respectively. The filament feeding assembly has first and second suspensions arranged to adjustably couple the respective first and second connectors to the respective first and second guide rails. The first and second connectors pass each other by way of adjusting the distance between the first connector and the first guide rail and/or adjusting the distance between the second connector and the second guide rail.

10 Claims, 16 Drawing Sheets



(51) **Int. Cl.**
B29C 64/209 (2017.01)
B29C 64/393 (2017.01)
B33Y 30/00 (2015.01)
B33Y 50/02 (2015.01)

(52) **U.S. Cl.**
CPC ***B29C 64/393*** (2017.08); ***B33Y 30/00*** (2014.12); ***B33Y 50/02*** (2014.12)

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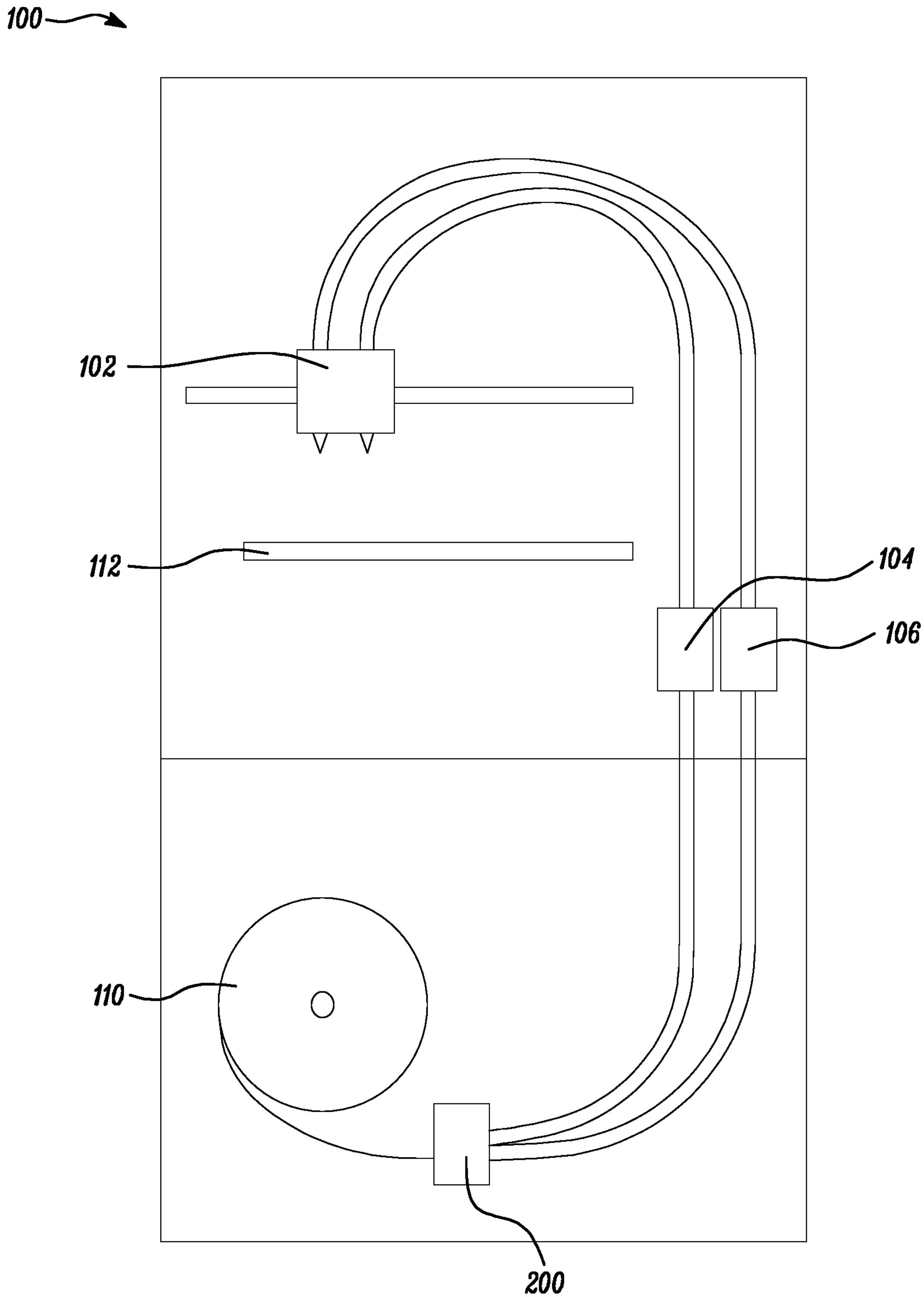
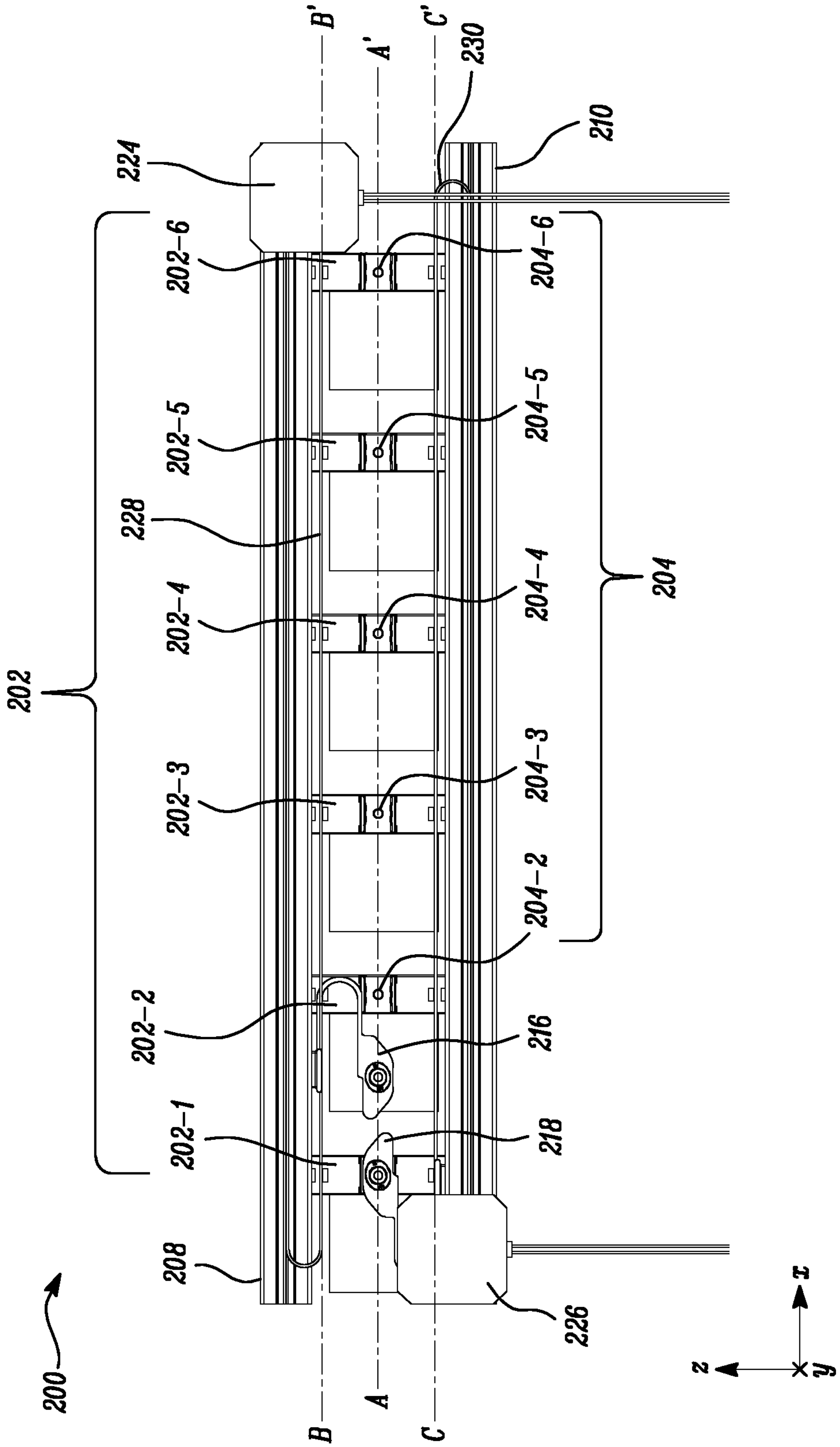


FIG. 1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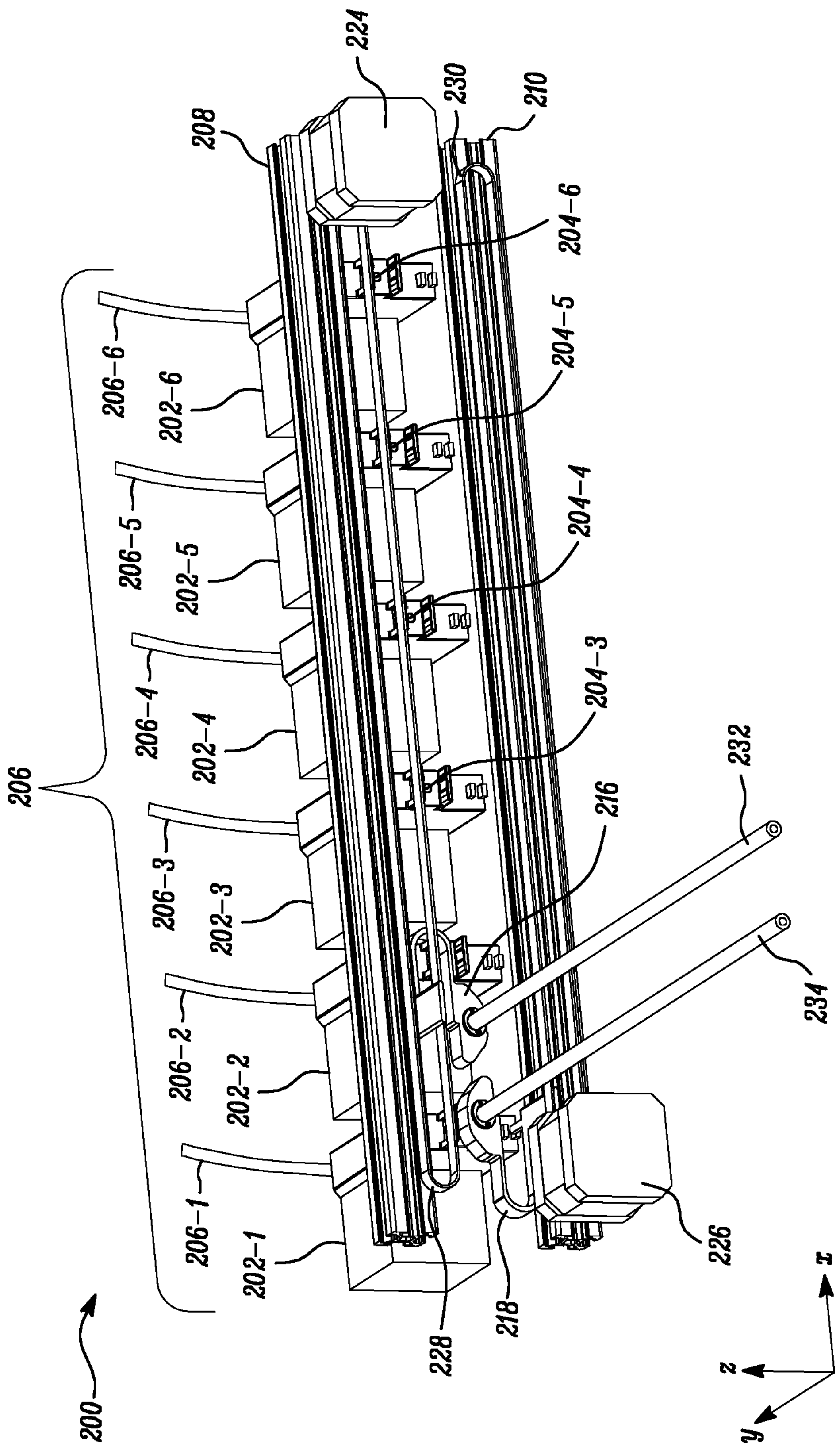
202
202-1
202-2
202-3
202-4
202-5
202-6
204
204-2
204-3
204-4
204-5
204-6
208
210
216
218
224
226
228
230
A
A'
B
B'
C
C'
x
y
z

FIG. 2A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230
210
208
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204-5
206-5
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204-4
206-4
202-4
204-3
206-3
202-3
216
232
206-2
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234
206-1
202-1
228
218
226
200
x
z
y

FIG. 2B

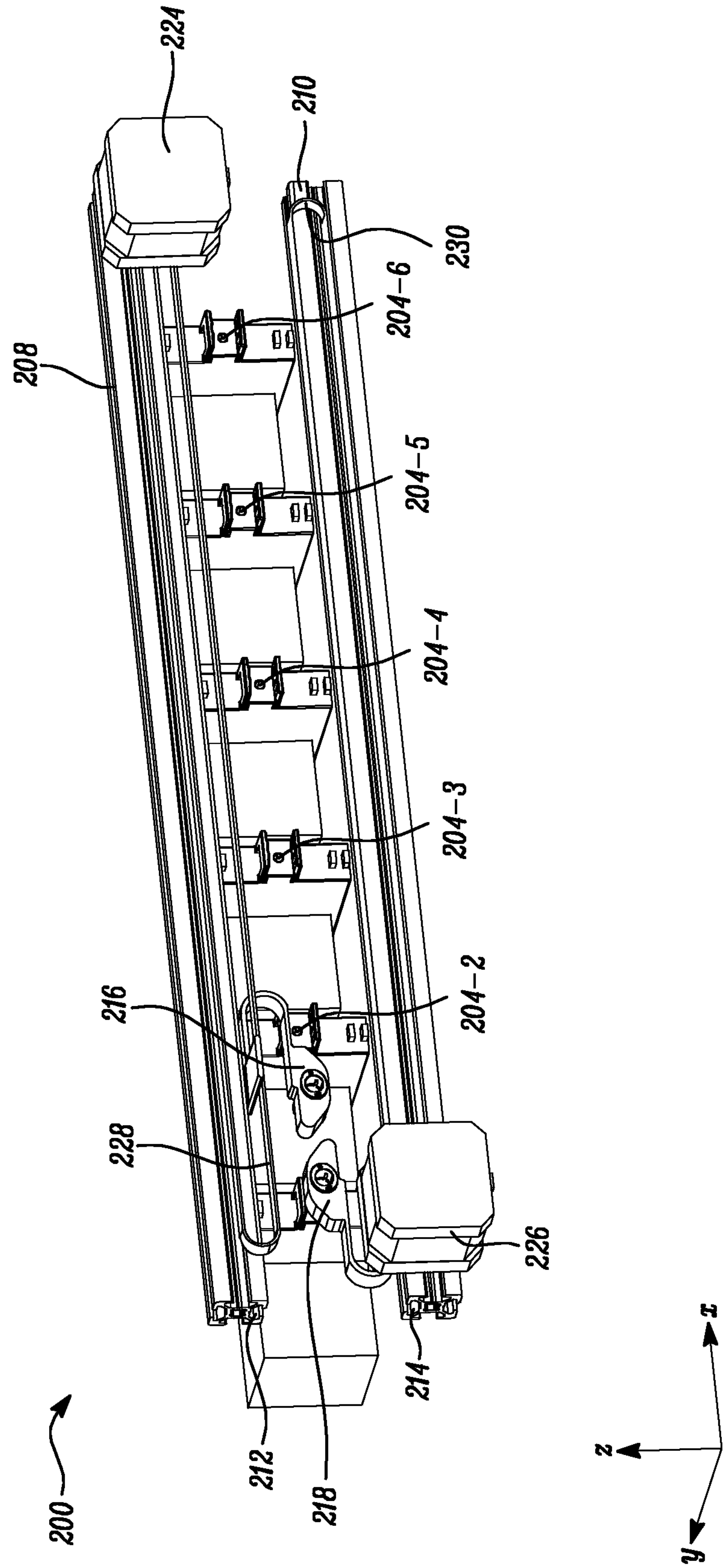

224
210
230
208
204-6
204-5
204-4
204-3
204-2
216
228
226
214
212
218
200
x
z
y

FIG. 2C

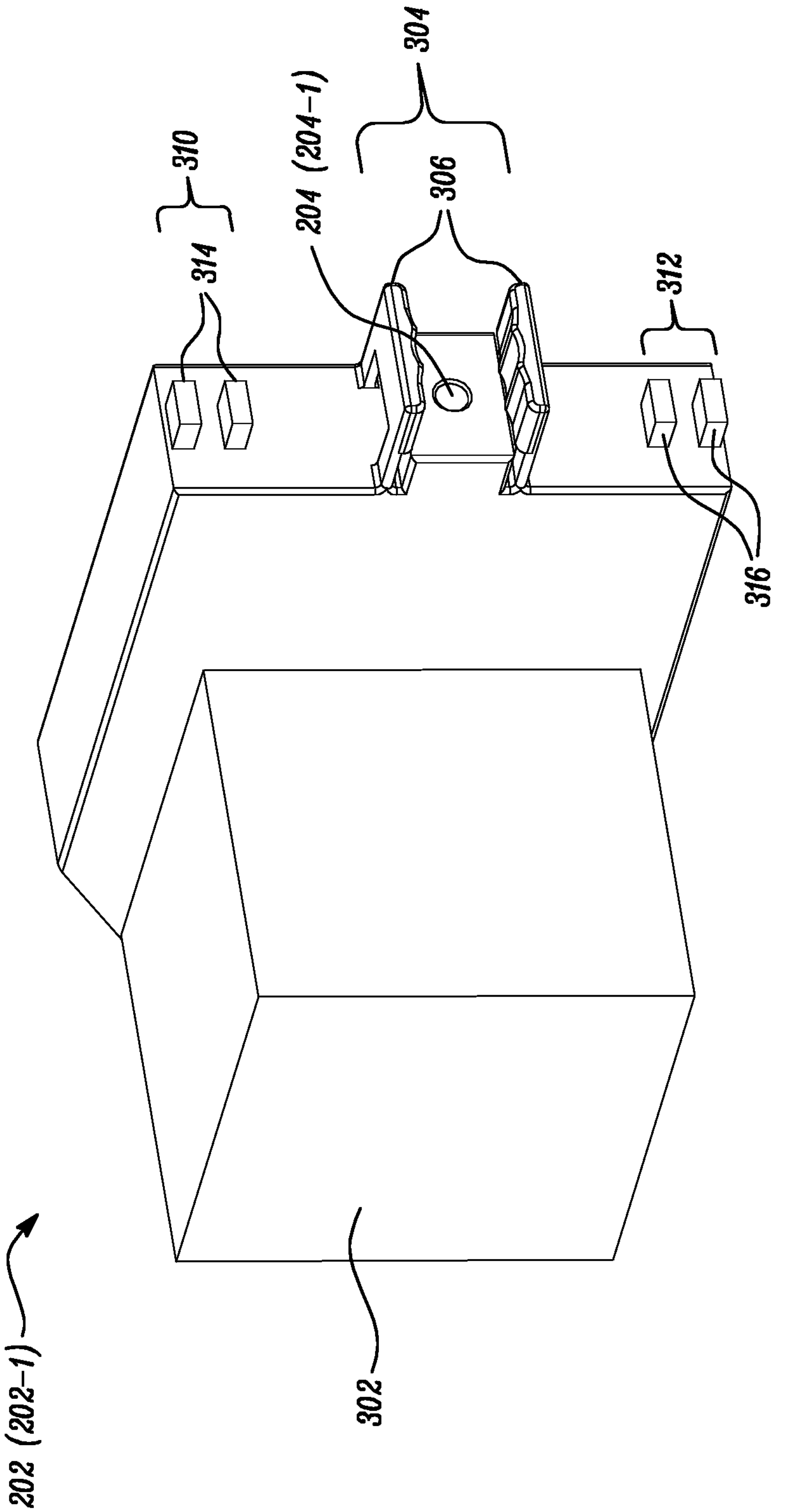
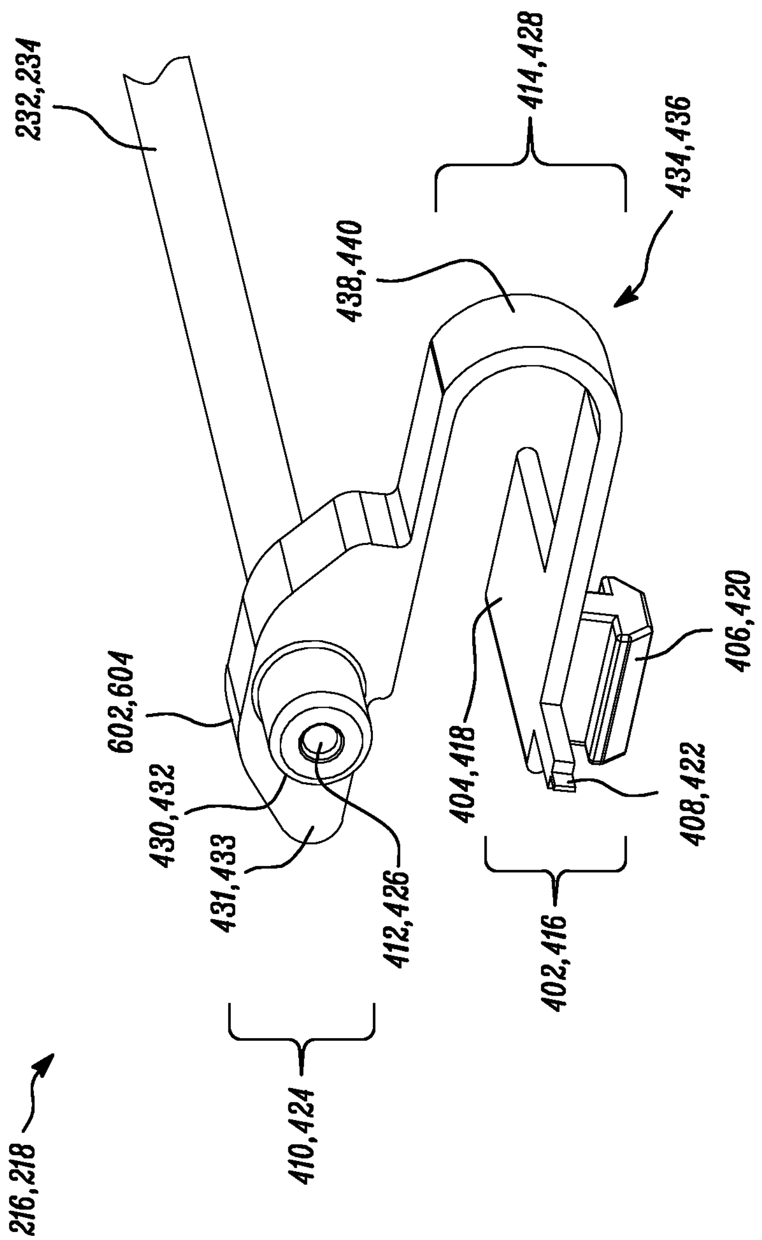


FIG. 3


232,234
414,428
434,436
438,440
406,420
404,418
408,422
402,416
602,604
430,432
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410,424
216,218

FIG. 4

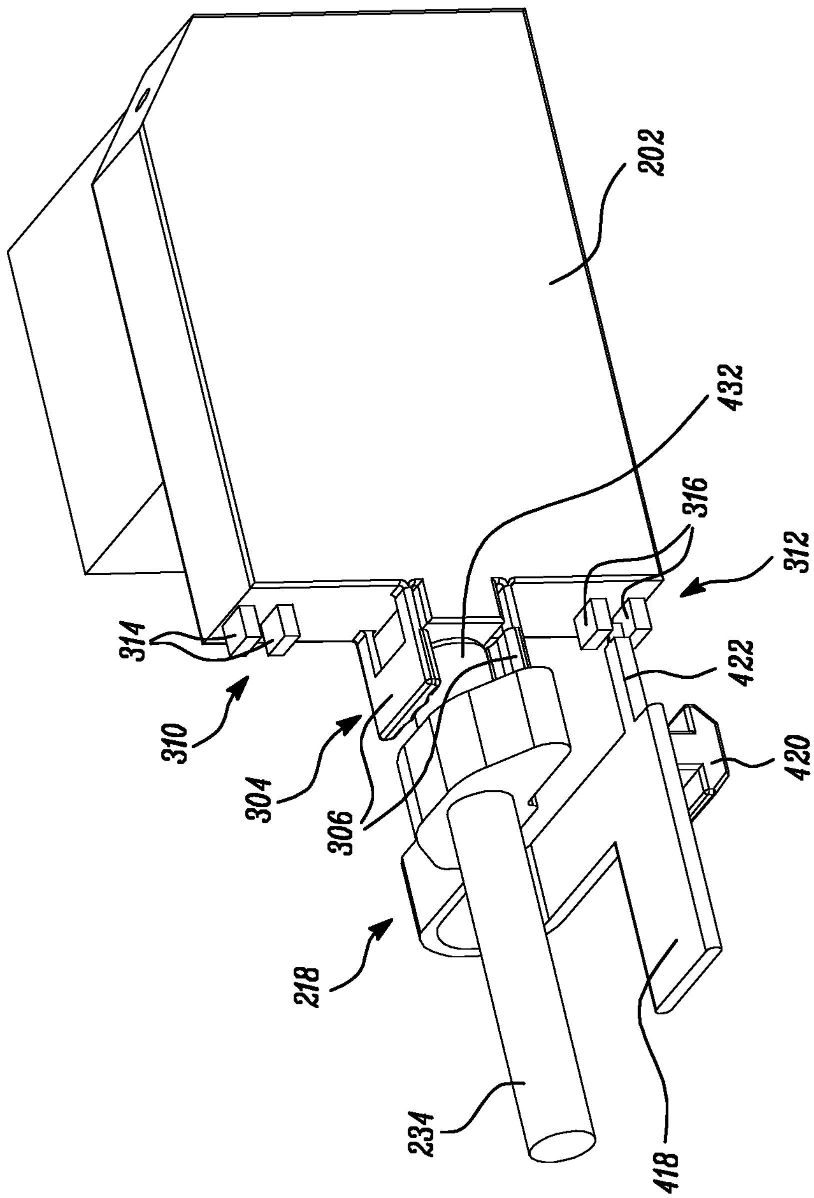


FIG. 5

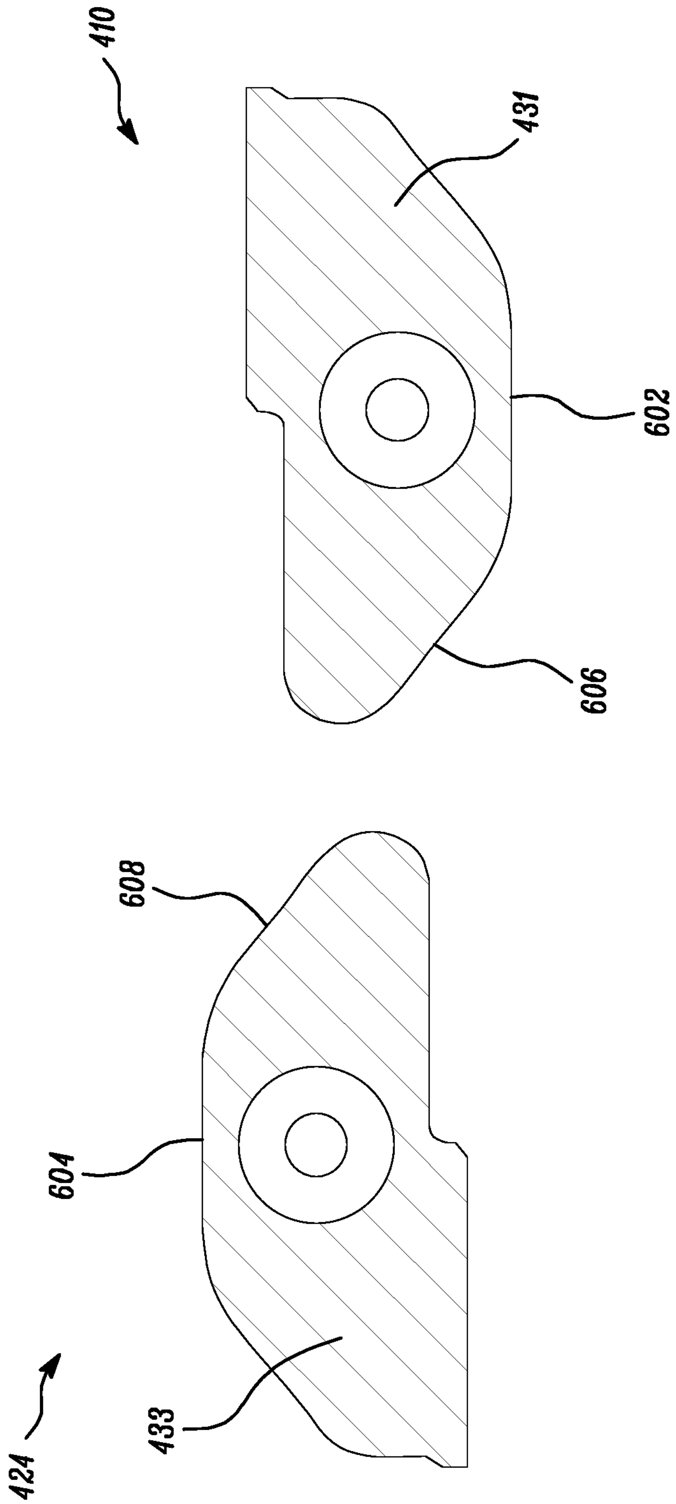


FIG. 6A

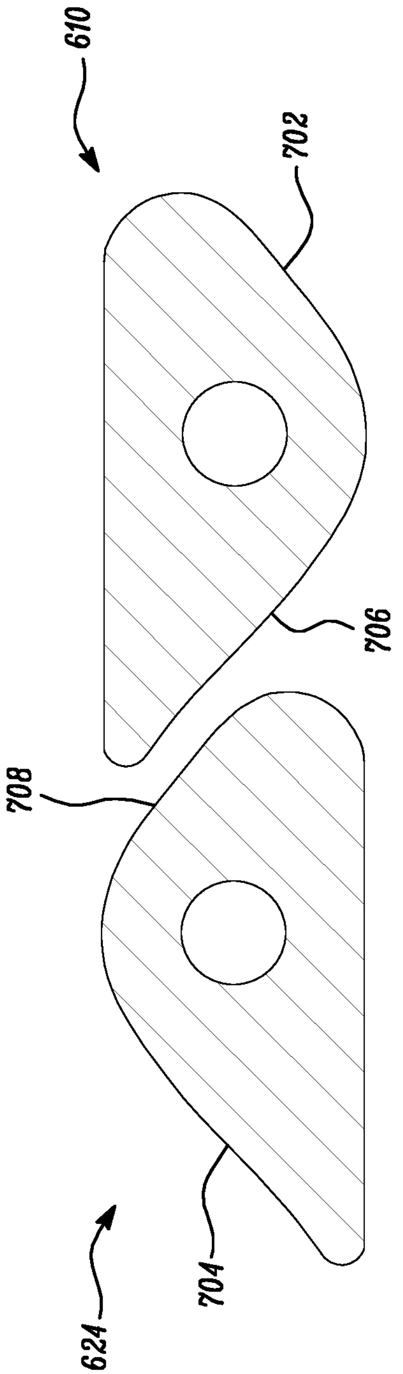


FIG. 6B

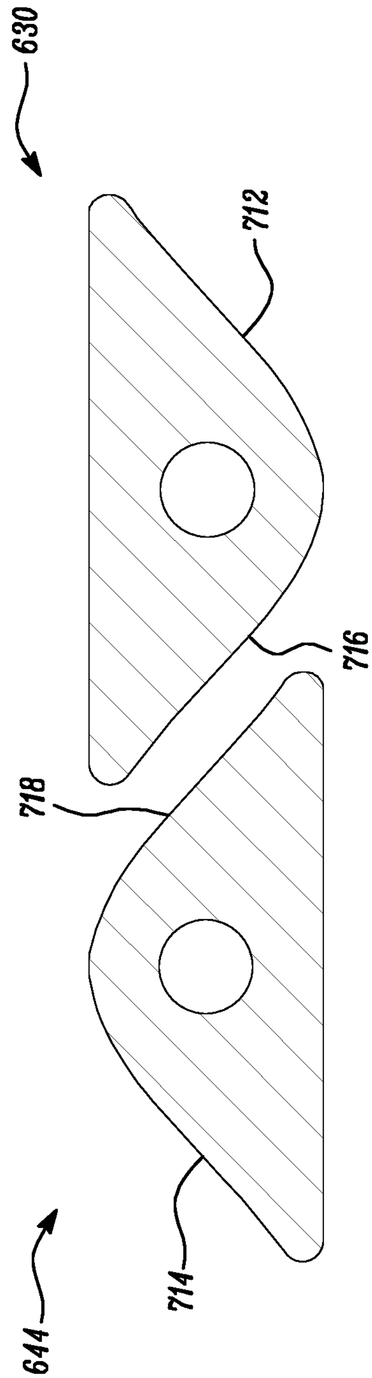
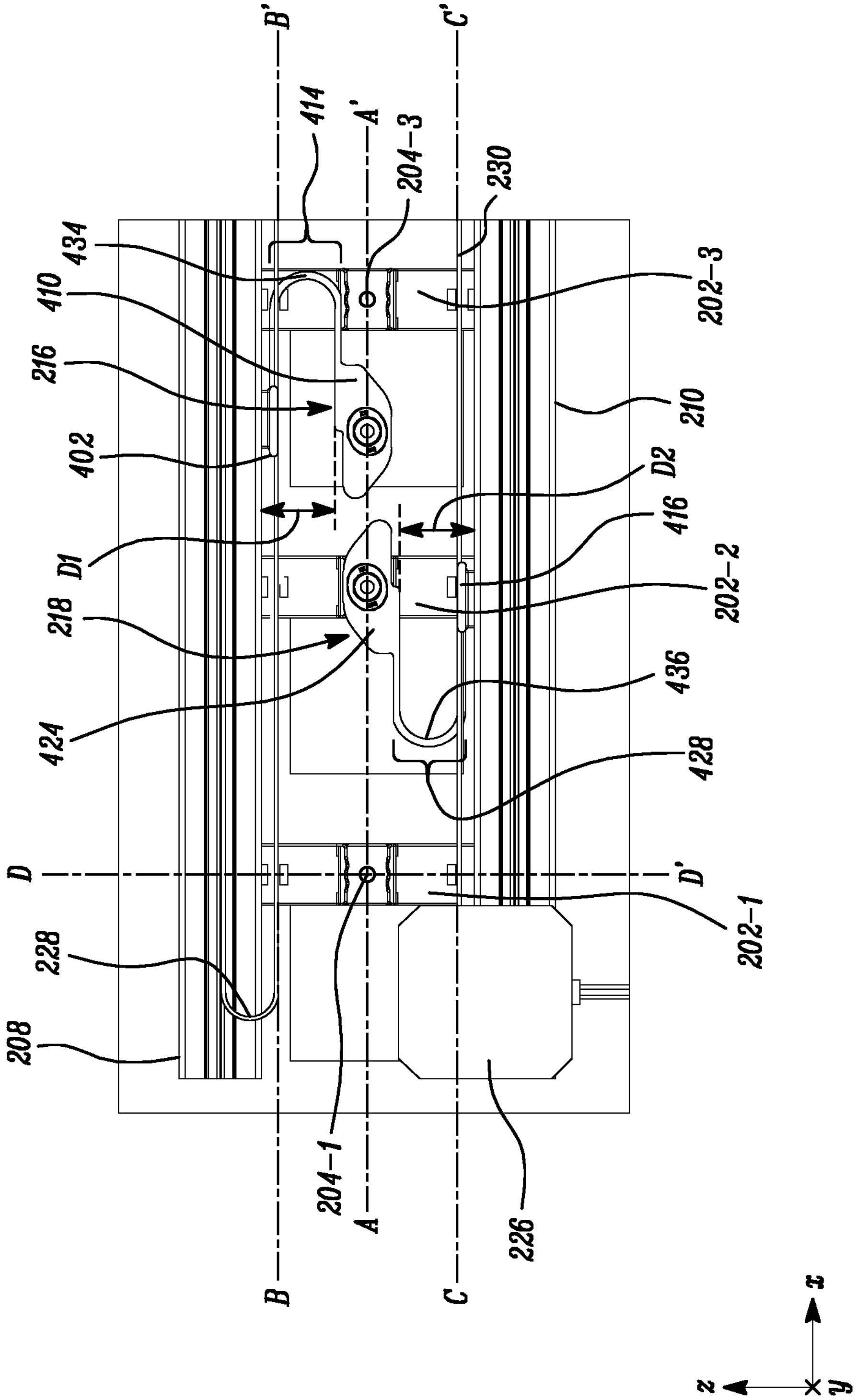


FIG. 6C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C'
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A'
204-3
230
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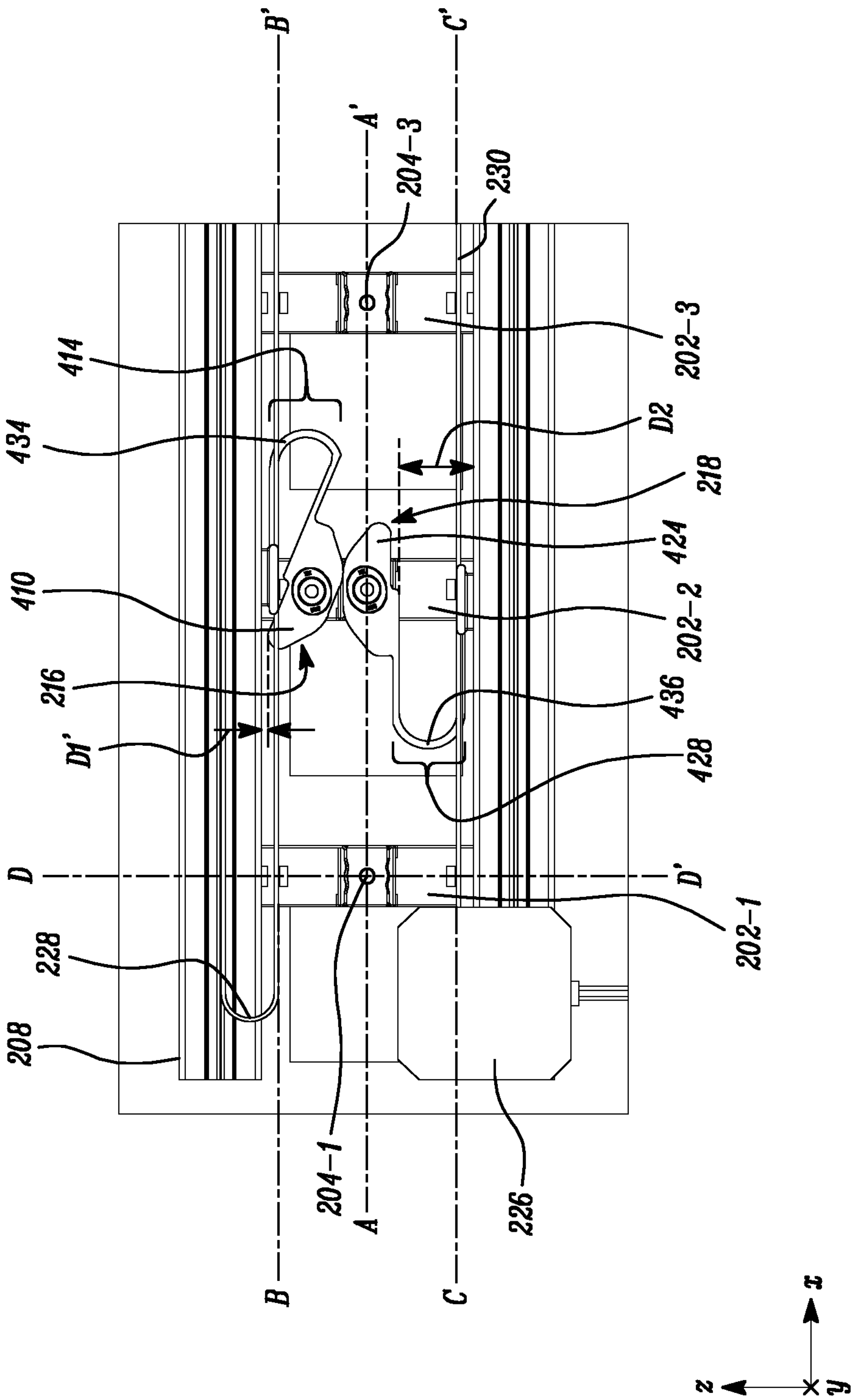
410
216
402
202-3
210
D2
416
D1
218
202-2
424
436
428
D
D'
228
208
202-1
204-1
A
B
C
226
x
z
y

FIG. 7A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A'
C'
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202-3
434
D2
218
424
410
202-2
216
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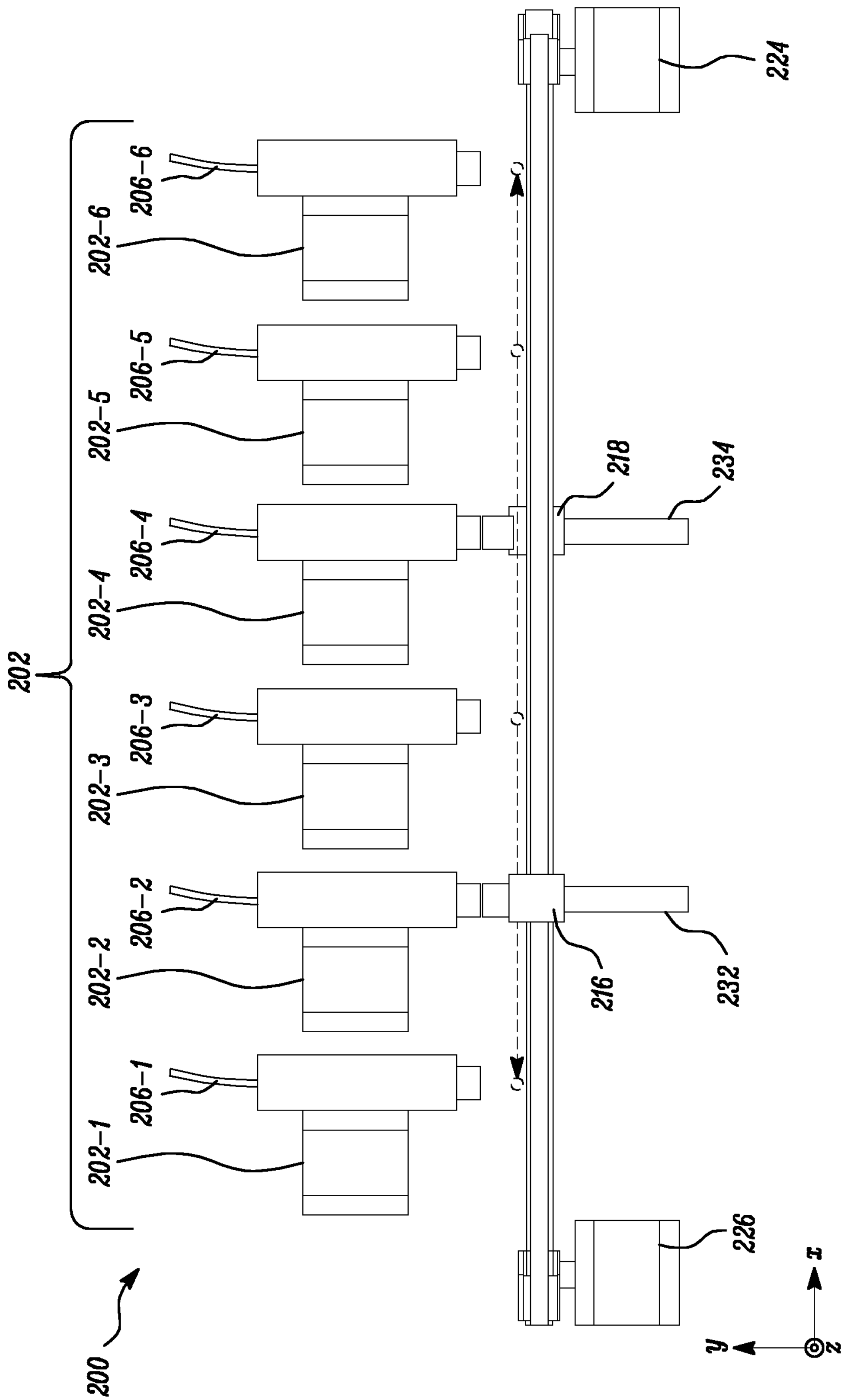
D1'
428
D
D'
228
202-1
208
204-1
A
226
B
C
x
z
y

FIG. 7B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202
202-1
202-2
202-3
202-4
202-5
202-6
206-1
206-2
206-3
206-4
206-5
206-6
216
218
224
226
232
234
x
y
z

FIG. 8

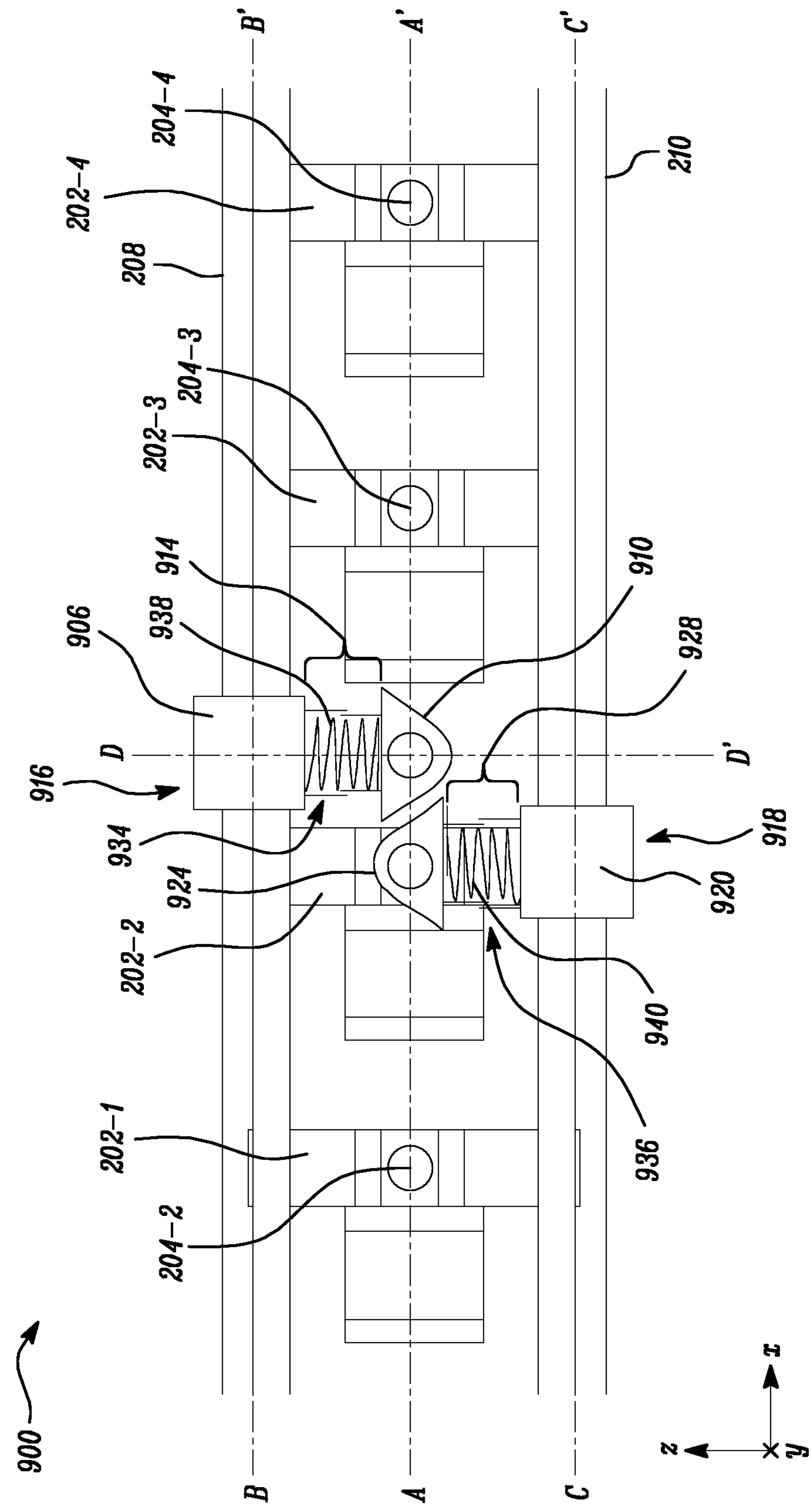
900
B
A
C
B'
A'
C'
D
D'
202-1
202-2
202-3
202-4
204-2
204-3
204-4
208
210
906
910
914
916
918
920
924
928
934
936
938
940
z
x
y

FIG. 9

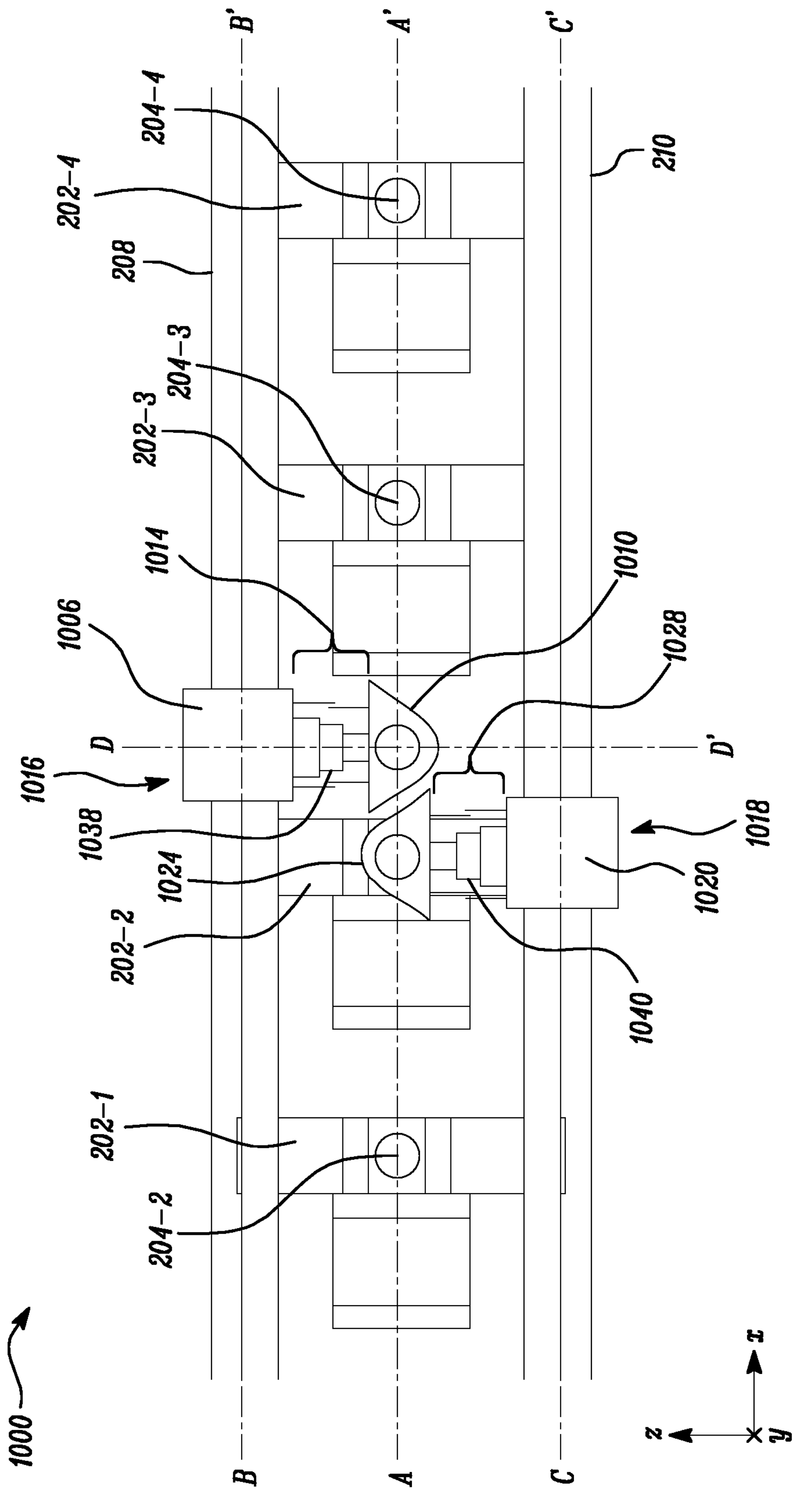
1000
B
B'
A
A'
C
C'
D
D'
202-1
204-2
202-2
1024
1038
1016
1006
1014
202-3
204-3
202-4
204-4
208
210
1010
1028
1018
1020
1040
z
x
y

FIG. 10

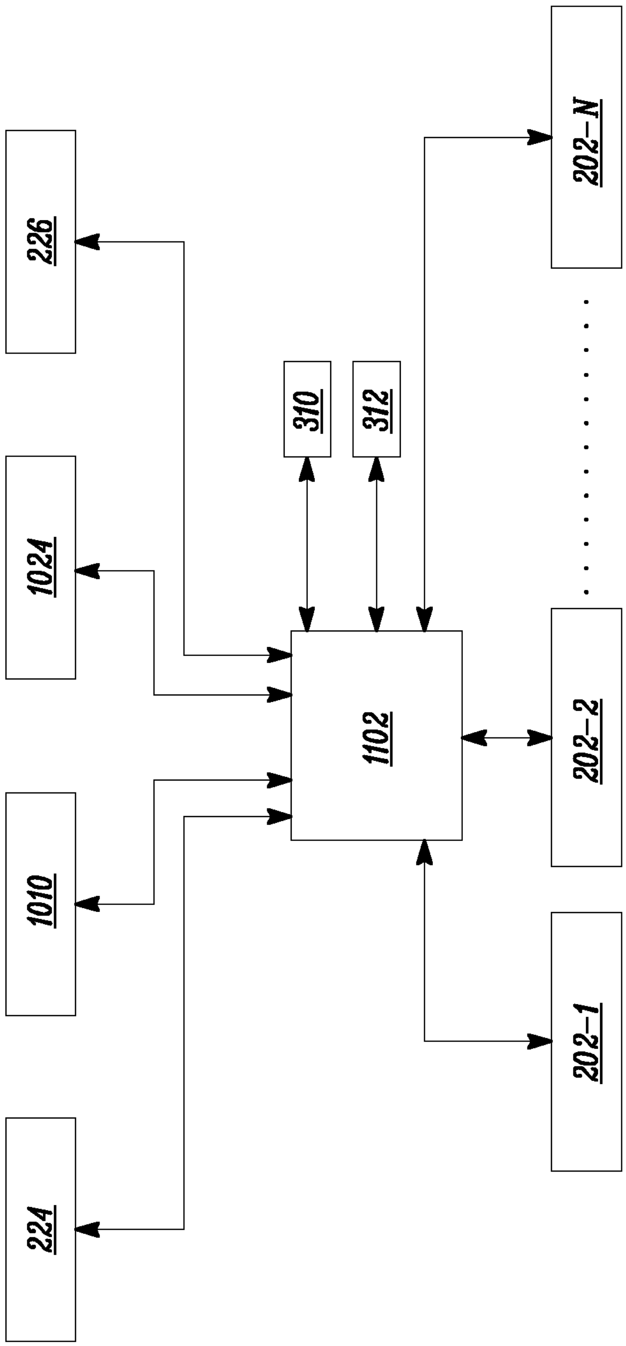

FIG. 11

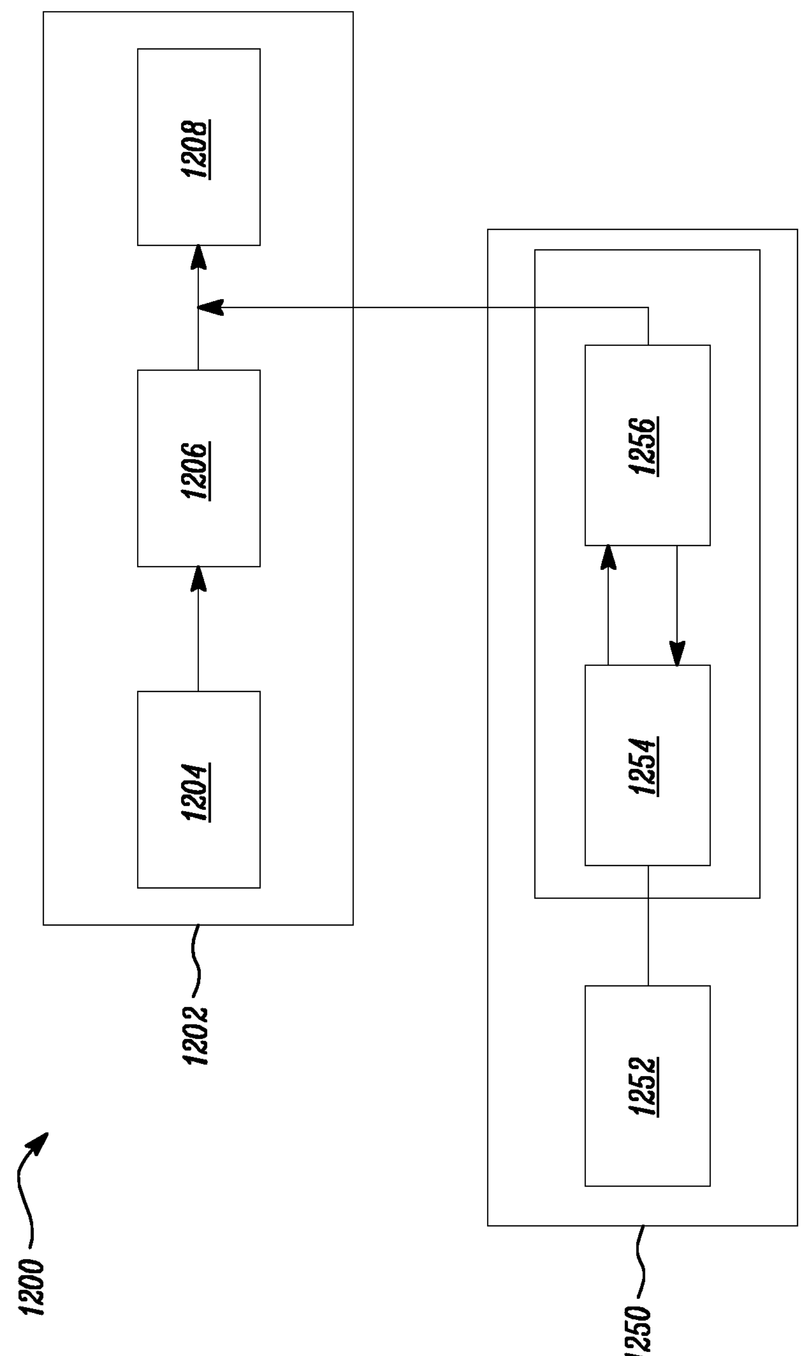

FIG. 12

FILAMENT FEEDING ASSEMBLY FOR FUSED FILAMENT FABRICATION SYSTEM

FIELD OF THE INVENTION

The present invention relates to a filament feeding assembly for a fused filament fabrication (FFF) system. It also relates to a fused filament fabrication system comprising such a filament feeding assembly.

BACKGROUND ART

Fused filament fabrication (FFF) is an additive manufacturing process that typically uses a continuous filament of a thermoplastic material. The filament may be fed from a filament supply to a moving, heated print head, and may be deposited through a print nozzle onto an upper surface of a build plate. Further, the print head may be moved relative to the build plate under computer control to define a printed shape. In certain FFF systems, the print head may move in two dimensions to deposit one horizontal plane, or a layer, at a time. A work is therefore formed by the deposited layers. The work or the print head may then be moved vertically by a small amount to begin a new layer. In this manner, a 3D-object may be produced out of the thermoplastic material.

Some print heads may utilize multiple extruders to deposit different thermoplastic materials or a combination thereof. The ability to extrude different thermoplastic materials may allow selection and use of different thermoplastic materials based on desired physical properties and/or geometry of the 3D-object. The different thermoplastic materials may also be selected based on intended applications. For example, some print heads may utilize a pair of extruders that may selectively extrude a part material and a support material. Using multiple extruders in a single printing system may require appropriate selection of a filament to be fed to the extruder that is operational at a given time.

U.S. Pat. No. 9,669,586 discloses a material dispensing system. The material dispensing system comprises a single drive motor attached to an additive manufacturing system via a motor mount. The material dispensing system further comprises a multi-material filament drive system, which is powered by the single drive motor via a flexible drive shaft. The filament drive system comprises a plurality of filament drives for a respective plurality of filaments. Rotary motion is transferred from the single drive motor to a selected one of filament drives. The motor enables the dispensing of an associated filament from the selected one of filament drives. The material dispensing system may be required to have a plurality of paths for the plurality of filaments dispensed by associated filament drives. This may increase the complexity of the material dispensing system. Further, the filaments are dispensed through separate filament drives, and thus feed force for each filament may be different, which may lead to uneven extrusion of filaments.

SUMMARY OF THE INVENTION

The aim of the present invention is to provide an improved filament feeding assembly for a fused filament fabrication (FFF) system.

According to a first aspect of the present invention, there is provided a filament feeding assembly for a fused filament fabrication (FFF) system comprising a plurality of filament feeders arranged next to each other in a row. Each filament feeder comprises an outlet. The respective outlets of the

filament feeders are aligned on a straight line. The filament feeding assembly further comprises a first guide rail and a second guide rail, both being arranged parallel to the straight line. The filament feeding assembly further comprises a first connector movably coupled to the first guide rail and comprising a first filament entrance for selectively receiving a filament from one of the outlets. The filament feeding assembly further comprises a second connector movably coupled to the second guide rail and comprising a second filament entrance for selectively receiving a filament from one of the outlets. The filament feeding assembly further comprises two actuators, each being arranged to move one of the connectors in front of the respective outlets of the filament feeders. The filament feeding assembly further comprises a first suspension arranged to adjustably couple the first connector to the first guide rail, so that a distance between the first guide rail and the first connector is adjustable. The filament feeding assembly further comprises a second suspension arranged to adjustably couple the second connector to the second guide rail, so that a distance between the second guide rail and the second connector is adjustable. The first and second connectors are arranged to pass each other by way of adjusting the distance between the first connector and the first guide rail and/or adjusting the distance between the second connector and the second guide rail.

The filament feeding assembly of the present invention may allow selective feeding of any two filaments from a plurality of filaments. The filament feeding assembly comprises two connectors that are movable in the filament feeding assembly and are arranged to move to and selectively engage with any of the plurality of feeders and receive respective filaments therefrom. Each of the connectors may be coupled to a Bowden tube that transports the filament received at the respective connector. Each of the connectors may be part of a so-called selector comprising a corresponding connector and a corresponding suspension. For example, a first selector comprises the first connector and the first suspension. Further, a second selector comprises the second connector and the second suspension.

The filament may be transported to a secondary feeder and then to a print head of the FFF system, or directly to the print head. In the case of a dual extruder print head of the FFF system, each connector feeds filaments to one of the extruders. The filament feeding assembly is arranged, such that filament is fed to one of the extruders of the print head through common channels, via one of the connectors. As a result, characteristics of the filament being fed to the print head, such as friction values and feed force values, are similar, irrespective of the feeder from where the filament is fed. This may allow the FFF system to extrude the filaments more reliably and more uniformly.

Further, since there is a single path for the filament to traverse, the FFF system comprising the proposed filament feeding assembly does not require a merger. This may further reduce friction that may otherwise occur due to movement of the filament through the merger. Specifically, in some cases, the proposed filament feeding assembly can be used without the merger. Hence, the path for the filament from the feeder to the print head may be reduced, which further reduces the friction for the filament and also leads to less wasted material. Further, moisture ingress into the filament path is limited.

Additionally, the filament feeding assembly may be automated. Since there are two connectors, the chances of filaments being fed in a wrong order is minimized.

In an embodiment, each of the first and second connectors comprises a wedge-shaped outer wall. Each of the first and second suspensions may comprise a resilient member. The resilient member is arranged to resiliently deform when the outer wall of the first connector pushes against the outer wall of the second connector.

In an embodiment, the resilient member comprises a curved member fixedly coupled to the respective connector. Such a curved member can be an integral part with the body of the connector and is easy to manufacture and requires little or no maintenance. The curved member may be manufactured using a plastic or other flexible material such as a metal. It is noted that a thickness of the curved member is chosen so that the member is both strong and flexible as is appreciated by the skilled person.

In an embodiment, the resilient member comprises a coil spring. The coil spring compresses when the outer wall of one of the connectors pushes against the outer wall of the other connector. The spring coils are passive elements and are compressed due to force acting on respective selectors.

In an embodiment, at least one of the first and second suspensions comprises a telescoping mechanism. The telescoping mechanisms can be actuated using any suitable actuator. This may allow active and accurate adjustment of the selectors.

In an embodiment, each of the filament feeders comprises a dock. Each of the connectors may comprise a protrusion arranged to be received in the dock of the filament feeder. The dock restricts motion of the protrusion, and consequently, the connector as such, once the protrusion is received in the dock. This may allow filament to be fed from the feeder to the connector smoothly without any undesired relative movement between the connector and the filament feeder in a coupled state.

In an embodiment, each of the filament feeders comprises a sensor arranged to generate a signal indicative of an engagement of one of the connectors with the filament feeder. The filament feeding assembly may be arranged such that a filament feeder begins feeding the corresponding filament only after the signal is generated indicative of the engagement of one of the connectors with the filament feeder.

In an embodiment, the filament feeding assembly further comprises a controller arranged to determine target feeders from the plurality of filament feeders for the connectors. The controller is arranged to control the actuators to move each of the associated connectors to one of the target feeders. The controller is then arranged to determine that the connectors are engaged with the target feeders. The controller is further arranged to operate the target feeders to feed filament through the connectors.

In an embodiment, the controller is further arranged to move at least the first connector towards the one of the target feeders. The controller is then arranged to determine that the first connector is proximal to the second connector. The controller is arranged to control the first suspension, such that the first connector is retracted towards the first guide rail, in order to allow the first connector to pass the second connector. The controller may be further arranged to control the first suspension, such that the first connector moves to a non-retracted state after the first connector has passed the second connector.

The use of the controller may allow accurate operation of the filament feeding assembly. Further, the controller may be provided with a transceiver unit that allows the controller to receive and send instructions wirelessly and remotely.

According to a second aspect of the present invention, there is provided a fused filament fabrication system comprising the filament feeding assembly of the first aspect. The proposed filament feeding assembly may provide a common path for the filaments to be extruded by the fused filament fabrication system, and thereby enables the fused filament fabrication system to extrude filaments uniformly.

BRIEF DESCRIPTION OF THE DRAWINGS

These and other aspects of the invention are apparent from and will be elucidated with reference to the embodiments described hereinafter. In the drawings,

These and other aspects of the invention are apparent from and will be elucidated with reference to the embodiments described hereinafter. In the drawings,

FIG. **1** schematically shows a side view of a fused filament fabrication (FFF) system, according to an embodiment of the present invention;

FIG. **2**A schematically shows a front view of a filament feeding assembly of the FFF system, according to an embodiment of the present invention;

FIG. **2**B schematically shows a top perspective view of the filament feeding assembly, according to an embodiment of the present invention;

FIG. **2**C schematically shows a side perspective view of the filament feeding assembly, according to an embodiment of the present invention;

FIG. **3** schematically shows a perspective view of a filament feeder of the filament feeding assembly, according to an embodiment of the present invention;

FIG. **4** schematically shows a perspective view of a selector of the filament feeding assembly, according to an embodiment of the present invention;

FIG. **5** schematically shows a perspective view an engagement between the selector and the filament feeder, according to an embodiment of the present invention;

FIG. **6**A schematically shows cross-sectional views of first and second connectors of the filament feeding assembly, according to an embodiment of the present invention;

FIG. **6**B schematically shows cross-sectional views of the first and second connectors, according to another embodiment of the present invention;

FIG. **6**C schematically shows cross-sectional views of the first and second connectors, according to another embodiment of the present invention;

FIG. **7**A schematically shows a front view of a part of the filament feeding assembly when the first and second selectors are in respective normal states, according to an embodiment of the present invention;

FIG. **7**B schematically shows a front view of a part of the filament feeding assembly when the first and second selectors are passing each other, according to an embodiment of the present invention;

FIG. **8** schematically shows a top view of another filament feeding assembly, according to an embodiment of the present invention;

FIG. **9** schematically shows a front view of another filament feeding assembly, according to an embodiment of the present invention;

FIG. **10** schematically shows a front view of another filament feeding assembly, according to an embodiment of the present invention;

FIG. **11** schematically shows a controller of the filament feeding assembly, according to an embodiment of the present invention; and

FIG. **12** schematically shows a state diagram for one of the filament feeders of the filament feeding assembly.

It should be noted that items which have the same reference numbers in different FIGURES, have the same structural features and the same functions, or are the same signals. Where the function and/or structure of such an item has been explained, there is no necessity for repeated explanation thereof in the detailed description.

DETAILED DESCRIPTION OF EMBODIMENTS

In a fused filament fabrication (FFF) process, a strand or filament of a material, such as a thermoplastic material, is forced through a heated extruder nozzle, which is arranged and controlled to deposit layers of molten filament on a print bed.

FIG. **1** schematically shows a side view of a fused filament fabrication (FFF) system **100**, according to an embodiment of the present invention. The FFF system **100** comprises a dual extruder print head **102**. The print head **102** is fed respective filaments from two feeders **104**, **106**. The two feeders **104**, **106** are, in turn, fed filament from a filament feeding assembly **200**. The filament feeding assembly **200** is arranged to receive filament from a number of filament spools **110**. Once filament is loaded from the feeders **104**, **106** via the filament feeding assembly **200**, the print head **102** is able to deposit extruded filament onto a print bed **112**. A dual extruder print head, also referred to as dual nozzle print head, provides many advantages to the user as compared to a single nozzle print head, as will be appreciated by the skilled person.

It is noted that the feeders **104**, **106** may be arranged within or on the print head **102** resulting in a so-called direct drive printing system. Furthermore, it is conceivable that the feeders **104**, **106** are absent and that the filament feeding assembly **200** is directly controlling a filament feeding speed to the print head **102**.

FIGS. **2**A-**2**C schematically show a front view, a top perspective view, and a side perspective view, respectively, of a specific filament feeding assembly **200** of the FFF system **100**. The filament feeding assembly **200** is interchangeably referred to as "the assembly **200**". In FIGS. **2**A-**2**C mutually orthogonal x, y and z-axes are indicated. In this example, the x and y-axes are disposed along a horizontal plane of the assembly **200** and the z-axis is disposed orthogonal to the horizontal plane (i.e., x-y plane) of the assembly **200**. It should however be noted that the orientation of the assembly **200** within the FFF system can be different from the orientation shown in the figures.

The assembly **200** comprises a plurality of filament feeders **202**. The filament feeders **202** are interchangeably referred to as "feeders **202**". In some embodiments, the filament feeders **202** are referred to as pre-feeders since they are arranged upstream from the feeders **104**, **106** shown in FIG. **1**. The plurality of filament feeders **202** are arranged next to each other in a row. Each filament feeder **202** comprises an outlet **204**. In other words, the plurality of feeders **202** comprises a respective plurality of outlets **204**. The plurality of feeders **202** are arranged such that the respective outlets **204** of the feeders **202** are aligned on a straight line A-A'. In the illustrated embodiment of FIGS. **2**A-**2**C, the straight line A-A' is parallel to the x-axis.

The plurality of feeders **202** is arranged to selectively dispense respective plurality of filaments **206**, see FIG. **2**B. The plurality of filaments **206** are dispensed through the respective outlets **204**. The plurality of filaments **206** generally comprise thermoplastic materials and may initially be stored on a spool. The filaments **206** may be of different types, each exhibiting different characteristics. In some embodiments, each feeder **202** from the plurality of feeders **202** dispenses respective filaments **206** that may be of a same type. In some other embodiments, each feeder **202** from the plurality of feeders **202** dispenses respective filaments **206** that may be of different types.

In the illustrated embodiment of FIGS. **2**A-**2**C, the assembly **200** comprises six filament feeders **202-1**, **202-2**, **202-3**, **202-4**, **202-5**, **202-6** (the filament feeders are collectively referred to by the numeral reference **202**). The number of filament feeders may be different and depends on the requirements. The plurality of feeders **202** is arranged to selectively dispense respective filaments **206-1**, **206-2**, **206-3**, **206-4**, **206-5**, **206-6** (the filaments are collectively referred to by the numeral reference **206**) through their respective outlets **204-1** (shown in FIG. **3**), **204-2**, **204-3**, **204-4**, **204-5**, **204-6** (the respective outlets are collectively referred to by the numeral reference **204**).

The assembly **200** further comprises a first guide rail **208** and a second guide rail **210**, both being arranged parallel to the straight line A-A'. In some embodiments, the first and second guide rails **208**, **210** are arranged, such that the straight line A-A' is positioned between the first and second guide rails **208**, **210**. In some embodiments, the first guide rail **208** is arranged along a first axis B-B' and the second guide rail **210** is arranged along a second axis C-C', see also FIG. **2**A. Therefore, the first axis B-B' and the second axis C-C' are both parallel to the straight line A-A'. Further, the first axis B-B' and the second axis C-C' are positioned such that the straight line A-A' is between the first axis B-B' and the second axis C-C'.

As is indicated in FIG. **2**C, in this embodiment, the first and second guide rails **208**, **210** define first and second channels **212**, **214** respectively, extending at least partially along their respective lengths. Each of the first and second channels **212**, **214** may have any suitable shape, for example, C-shaped, U-shaped, or any other shape based on application requirements.

In the embodiment of FIGS. **2**A-**2**C, the assembly **200** further comprises a first and a second selector **216**, **218**. The first selector **216** is movably coupled to the first guide rail **208**, and the second selector **218** is movably coupled to the second guide rail **210**. The first and second selectors **216**, **218** are movable relative to the plurality of feeders **202**. The first and second selectors **216**, **218** are movably coupled to the first and second guide rails **208**, **210**, respectively, such that the first and second selectors **216**, **218** are movable at least partially along lengths of the first and second guide rails **208**, **210**, respectively. In other words, the first selector **216** is movable along the first axis B-B', and the second selector **218** is movable along the second axis C-C'. The first and second selectors **216**, **218** are each arranged to engage with any one of the plurality of feeders **202**, such that, upon engagement, the first and second selectors **216**, **218** are arranged to receive the filament **206** from the respective feeders **202**. In this embodiment, the first and second selectors **216**, **218** are arranged to be at least partially and slidably received in the first and second channels **212**, **214**, respectively.

The assembly **200** further comprises two actuators **224**, **226**, each being arranged to move one of the selectors **216**, **218** in front of the respective outlets **204** of the feeders **202**. Specifically, each of the actuators **224**, **226** is arranged to move an associated selector **216**, **218** in front of the respective outlets **204** of the feeders **202**. The two actuators **224**, **226** are interchangeably referred to as first and second actuators **224**, **226**. In other words, the first and second actuators **224**, **226** are arranged to move connectors **410**, **424** (see also FIG. **4**) of the first and second selectors **216**, **218** respectively, in front of the respective outlets **204** of the feeders **202**. In some embodiments, each of the first and second actuators **224**, **226** may comprise stepper motors, dc motors, ac motors, and the like.

In this embodiment, the assembly **200** further comprises first and second drive mechanisms **228**, **230** arranged to drivably couple the first and second selectors **216**, **218** with the first and second actuators **224**, **226**, respectively. In other words, the first and second actuators **224**, **226** effect movement of the first and second selectors **216**, **218** through the first and second drive mechanisms **228**, **230**, respectively. In some embodiments, each of the first and second drive mechanisms **228**, **230** may comprise a belt drive, a chain drive, a gear drive, a friction drive, or combinations thereof.

As can be seen from, e.g., FIG. **2**B, the first and second selectors **216**, **218** are inverted relative to each other, such that first and second guide couplers **402**, **416** (see also FIG. **4**) can each be coupled to different guide rails **208**, **210** and driven by the associated drive mechanisms **228**, **230**.

In the embodiment of FIG. **2**B, the first and second selectors **216**, **218** are coupled to first and second Bowden tubes **232**, **234**, respectively. The first and second Bowden tubes **232**, **234** are flexible hollow tubes that transport the respective filaments **206** from the first and second selectors **216**, **218** towards the feeders **104**, **106** (shown in FIG. **1**).

FIG. **3** schematically shows a perspective view of one of the filament feeders **202** of the filament feeding assembly **200**. Each of the plurality of feeders **202** may comprise a drive mechanism **302** such that, on actuation, it dispenses the respective filament **206** (shown in FIG. **2**B) through the associated outlet **204** of the feeder **202**. The drive mechanism **302** may comprise a motor such as a stepper motor, a dc motor, an ac motor, and the like. In case the filament feeder **202** corresponds to the filament feeder **202-1** (shown in FIG. **2**A), the outlet **204** corresponds to the outlet **204-1**.

In some embodiments, each of the plurality of feeders **202** comprises a dock **304**. In some embodiments, the dock **304** comprises a pair of docking portions **306** spaced apart from each other. In this example, the pair of docking portions **306** are two protruding walls arranged to receive and selectively engage with protrusions **430**, **432** of the connectors **410**, **424** which will be discussed in more detail with reference to FIG. **4**. Once the respective protrusion **430**, **432** is engaged with the pair of docking portions **306** of the feeder **202**, the respective connector **410**, **424** is ready to receive the filament **206** dispensed from the outlet **204**. In this example, the pair of docking portions **306** are arranged around the outlet **204** of the respective feeder **202**, but alternatively the pair of docking portions **306** could be arranged remote from the outlet **204** to receive another part of the selectors **216**, **218**, depending on a configuration of the filament feeding assembly **200**.

In the illustrated embodiment of FIG. **3**, each of the filament feeders **202** comprises first and second sensors **310**, **312** arranged to generate signals indicative of the engagement of the first and second connectors **410**, **424** respectively, with the respective feeder **202**. In some embodiments, the first sensor **310** comprises a first pair of sensor elements **314**. Further, the second sensor **312** comprises a second pair of sensor elements **316**. The first and second sensors **310**, **312** may be contactless sensors arranged to detect the presence of a flag (e.g., first and second flags **408**, **422**, see also FIG. **4**) of one of the first and second selectors **216**, **218** as will be discussed with reference to FIG. **5**.

FIG. **4** schematically shows a perspective view of any one the first and second selectors **216**, **218** of the filament feeding assembly **200**. The first selector **216** comprises the first connector **410**, a first suspension **414** and the first guide coupler **402**. The first guide coupler **402** is movable by the first actuator **224** through the first drive mechanism **228**, see also FIG. **2**B. Consequently, the whole first selector **216** is movable by means of the first actuator **224** and the first drive mechanism **228**. In this embodiment, the first guide coupler **402** comprises a first drive connection **404** and a first slide **406**. The first drive connection **404** is arranged to be connected with the first drive mechanism **228**. In case the first drive mechanism **228** comprises a belt, the first drive connection **404** is a belt connection arranged to be driven by the belt. Thus, in some embodiments, when the first actuator **224** drives the first drive mechanism **228**, the first drive connection **404** is subsequently driven, thereby effecting movement of the first selector **216**.

The first slide **406** is arranged to be at least partially and slidably received within the first channel **212** of the first guide rail **208**. In other words, when the first selector **216** is driven by the first drive mechanism **228**, the first slide **406** slides within the first channel **212**. Further, the first slide **406** slidably supports the first selector **216** on the first guide rail **208**. In some embodiments, the first slide **406** may have a cross-sectional geometry that matches a cross-sectional geometry of the first channel **212**, such that the first slide **406** slidably fits in the first channel **212**. In some embodiments, the cross-sectional geometry of the first slide **406** may be wedge shaped. Further, the first channel **212** may be arranged to restrict movement of the first slide **406**, such that the first slide **406** moves substantially along the length of the first channel **212**. Thus, the movement of the first selector **216** is restricted by the movement of the first slide **406**. In some embodiments, the first slide **406** moves along the first axis B-B', and, consequently, the first selector **216** moves along the first axis B-B'. It is noted that alternatively, the first slide **406** could also comprise a linear bearing.

In this embodiment, the first selector **216** further comprises the first flag **408**. The first flag **408** is arranged to be at least partially received between the first pair of sensor elements **314** (shown in FIG. **3**) of the first sensor **310** of one of the feeders **202** when the first selector **216** is at a proximal location to the one feeder **202**, such that, at the proximal location, the first selector **216** is engaged with the feeder **202**. Further, the first sensor **310** is arranged to generate the signal indicative of the engagement of the first selector **216** with the feeder **202** when the respective first pair of sensor elements **314** at least partially receives the first flag **408** therebetween.

Similarly, the second selector **218** comprises the second connector **424**, a second suspension **428** and the second guide coupler **416**. The second guide coupler **416** further comprises a second drive connection **418** and a second slide **420**. The second selector **218** further comprises the second flag **422**.

The connectors **410**, **424** each comprises a filament entrance **412**, **426** for selectively receiving the filament **206** from one of the outlets **204**. The filament entrance **412**, **426** can receive the filament **206** when the respective connector **410**, **424** is positioned such that the filament entrance **412**, **426** is aligned with the respective outlet **204** of the feeder **202**. Further, the sensors **310**, **312** are arranged to generate a signal indicative of the engagement of the respective connectors **410**, **424** with the feeder **202** when the respective first pair of sensor elements **314** at least partially receives the flags **408**, **422** therebetween.

The first suspension **414** is arranged to adjustably couple the first connector **410** to the first guide rail **208**, so that a distance between the first guide rail **208** and the first connector **410** is decreased when the two connectors **410**, **424** need to pass each other.

The second suspension **428** is arranged to adjustably couple the second connector **424** to the second guide rail **210**, so that a distance between the second guide rail **210** and the second connector **424** is decreased. So, both connectors **410**, **424** move towards their respective guide rails **208**, **210** in case of a conflict (i.e., passing) of the connectors **410**, **424**.

It is noted that each feeder **202** can engage with only one of the first and second selectors **216**, **218** at a given time. Therefore, at a given time, the one feeder **202** engaged with the first selector **216** will be different from the one feeder **202** engaged with the second selector **218**. However, one of the feeders **202** may be engaged with both the first and second selectors **216**, **218** at different non-overlapping time periods.

In some embodiments, each of the first and second connectors **410**, **424** comprises a protrusion arranged to be received in the dock **304** (shown in FIG. **3**) of the filament feeder **202**. Specifically, the first and second connectors **410**, **424** comprise the first and second protrusions **430**, **432**, respectively. In the embodiment of FIG. **4**, the first and second protrusions **430**, **432** are disposed around the first and second filament entrances **412**, **426**, respectively. The first and second connectors **410**, **424** further comprise first and second main bodies **431**, **433**, respectively. The first and second main bodies **431**, **433** are connected to the first and second suspensions **414**, **428**, respectively. The first and second protrusions **430**, **432** extend from first and second main bodies **431**, **433**, respectively. Further, the first and second filament entrances **412**, **426** extend through the first and second protrusions **430**, **432** and the first and second main bodies **431**, **433**, respectively, such that the first and second filament entrances **412**, **426** communicate with the first and second Bowden tubes **232**, **234**. Each of the first and second protrusions **430**, **432** are arranged to be received in the dock **304** of the respective feeders **202**. Specifically, each of the first and second protrusions **430**, **432** are arranged to be received in the dock **304** of the respective feeders **202** when the first and second connectors **410**, **424** are engaged with the respective feeders **202**. When the first protrusion **430** of the first connector **410** is engaged with the dock **304** of the respective feeder **202**, the first filament entrance **412** of the first connector **410** is arranged to receive the filament **206** from the outlet **204** of the respective feeder **202**, through the first protrusion **430**. Similarly, when the second protrusion **432** of the first connector **424** is engaged with the dock **304** of the respective feeder **202**, the second filament entrance **426** of the second connector **424** is arranged to receive the filament **206** from the outlet **204** of the respective feeder **202**, through the second protrusion **432**.

In some embodiments, when each of the first and second protrusions **430**, **432** is received in the dock **304** of the respective filament feeder **202**, the dock **304** restricts movement of the corresponding one of the first and second protrusions **430**, **432** relative to the respective filament feeder **202**. Thus, when each of the first and second protrusions **430**, **432** is received in the dock **304** of the respective filament feeder **202**, the dock **304** restricts movement of the respective one of the first and second selectors **216**, **218** relative to the respective filament feeder **202**. This may ensure that, once any one of the first and second protrusions **430**, **432** is received in the dock **304** of the respective filament feeder **202**, the respective one of the first and second selectors **216**, **218** is held in place, such that the filament **206** from the respective filament feeder **202** is fed through the respective one of the first and second protrusions **430**, **432** without any disturbance or misalignment. Additionally or alternatively, the actuators **224**, **226** could be used to (actively) hold the selectors **216**, **218** in place when being engaged with one of the feeders **202**.

In an embodiment, each of the first and second suspensions **414**, **428** comprises a resilient member **434**, **436**, see FIG. **4**. The resilient members **434**, **436** are arranged to resiliently deform. When the resilient members **434**, **436** resiliently deform, the distance between the guide rails **208**, **210** and the respective connectors **410**, **424** changes.

In some embodiments, the resilient member **434**, **436** comprises a curved member fixedly coupled to the respective connector. FIG. **4** shows an example of resilient members **434**, **436** comprising first and second curved members **438**, **440**, respectively. The first and second curved members **438**, **440** are fixedly coupled to the first and second connectors **410**, **424**, respectively. The first and second curved members **438**, **440** connect the first and second connectors **410**, **424** to the first and second guide couplers **402**, **416**, respectively. The purpose of having flexible resilient members **434**, **436** will be discussed with reference to FIGS. **7A** and **7B**.

FIG. **5** schematically shows a perspective view of engagement of the second selector **218** with one of the feeders **202**. As shown, the second protrusion **432** is received by the dock **304** of the feeder **202**. Further, the second flag **422** is at least partially received between the second pair of sensor elements **316** of the second sensor **312**. In this engaged configuration, the second filament entrance **426**, see also FIG. **4**, can receive the respective filament **206** from the outlet **204** of the feeder **202**, see also FIG. **3**. The first selector **216** can be similarly engaged with one of the feeders **202**. In that case, the first protrusion **430** is received by the dock **304** of the feeder **202**. However, due to the positioning of the first selector **216** relative to the filament feeders **202**, the first flag **408** is at least partially received between the first pair of sensor elements **314** of the other sensor, i.e., the first sensor **310**.

FIG. **6**A schematically shows cross-sectional views of the first and second connectors **410**, **424**, according to an embodiment of the present invention. In the assembly **200** see also FIG. **2**A, the first and second selectors **216**, **218** are arranged such that the first and second connectors **410**, **424** are arranged in opposing orientations to one another. Referring to FIGS. **4** and **6**A, each of the first and second connectors **410**, **424** comprises an outer wall. Specifically, the first and second connectors **410**, **424** comprise first and second outer walls **602**, **604**, respectively. The first and second outer walls **602**, **604** are part of the first and second main bodies **431**, **433**, respectively. The first outer wall **602** comprises a first inclined surface **606** at one end. Similarly, the second outer wall **604** comprises a second inclined surface **608** at one end. The first and second inclined surface **606**, **608** are configured to selectively and slidingly engage with each other when the first and second connectors **410**, **424** pass each other. In the illustrated embodiment of FIG. **6**A, the first and second inclined surfaces **606**, **608** have substantially similar shapes. In some embodiments, each of the first and second outer walls **602**, **604** is wedge shaped. In other words, each of the first and second connectors **410**, **424** comprises wedge shaped outer wall **602**, **604**.

FIG. **6**B schematically shows cross-sectional views of first and second connectors **610**, **624**, according to another embodiment of the present invention. The first and second connectors **610**, **624** are functionally similar to the first and second connectors **410**, **424**, respectively. However, the first and second connectors **610**, **624** comprise first and second outer walls **702**, **704** having shapes that are different from that of the first and second outer walls **602**, **604** of the first and second connectors **410**, **424**, respectively. Further, the first outer wall **702** comprises a first inclined surface **706** at one end. Similarly, the second outer wall **704** comprises a second inclined surface **708** at one end. The first and second inclined surface **706**, **708** are configured to selectively and slidingly engage with each other when the first and second connectors **610**, **624** pass each other. In the illustrated embodiment of FIG. **6**B, the first and second inclined surfaces **706**, **708** have different shapes. The first and second outer walls **702**, **704** may be wedge shaped.

FIG. **6**C schematically shows cross-sectional views of first and second connectors **630**, **644** according to another embodiment of the present invention. The first and second connectors **630**, **644** are functionally similar to the first and second connectors **410**, **424**, respectively. However, the first and second connectors **630**, **644** comprise first and second outer walls **712**, **714** having shapes that are different from that of the first and second outer walls **602**, **604** of the first and second connectors **410**, **424**, respectively. In the illustrated embodiment of FIG. **6**C, each of the first and second outer walls **712**, **714** has an axis of symmetry passing through a center of the corresponding first and second connector **630**, **644**. Specifically, each of the first and second outer walls **712**, **714** may have a substantially symmetric triangular cross-sectional shape. Further, the first outer wall **712** comprises a first inclined surface **716** at one end. Similarly, the second outer wall **714** comprises a second inclined surface **718** at one end. The first and second inclined surface **716**, **718** are configured to selectively and slidingly engage with each other when the first and second connectors **630**, **644** pass each other. In the illustrated embodiment of FIG. **6**C, the first and second inclined surfaces **716**, **718** have substantially similar shapes.

FIG. **7**A schematically shows a front view of a part of the filament feeding assembly **200** when the first and second selectors **216**, **218** are in respective normal states. In the example of FIG. **7**A, the first selector **216** is not engaged with any of the feeders **202** whereas the second selector is engaged with the feeder **202-2**. In the normal state, the distance between the first connector **410** and the first guide rail **208** is equal to a distance D1. The distance D1 may correspond to a minimum distance between the first connector **410** and the first guide rail **208** along the z-axis when the first connector **410** is in the normal state. The distance D1 may further correspond to an offset between the straight line A-A' and the first axis B-B' along the z-axis. Further, in the normal state, the distance between the second connector **424** and the second guide rail **210** is equal to a distance D2. The distance D2 may correspond to a minimum distance between the second connector **424** and the second guide rail **210** along the z-axis when the second connector **424** is in the normal state. The distance D2 may further correspond to an offset between the straight line A-A' and the second axis C-C', along the z-axis.

FIG. **7**B schematically shows a front view of a part of the filament feeding assembly **200** when the first and second selectors **216**, **218** are in the process of passing each other. The first and second selectors **216**, **218** are arranged to pass each other by way of adjusting one of the first and second suspensions **414**, **428**. Specifically, the first and second connectors **410**, **424** are arranged to pass each other by way of adjusting the distance between the first connector **410** and the first guide rail **208** and/or adjusting the distance between the second connector **424** and the second guide rail **210**. In some embodiments, the resilient members **434**, **436** are arranged to resiliently deform along an adjustment axis D-D', where the adjustment axis D-D' is substantially along the z-axis.

In the illustrated embodiment of FIG. **7**B, the second selector **218** is engaged with the second feeder **202-2**. The first selector **216** is shown to be passing over the second selector **218**, as the first selector **216** is moving towards the first feeder **202-1**. In such a case, due to forces applied by the second connector **424** onto the first connector **410** (and vice versa), the first suspension **414** of the first selector **216** is adjusted such that the first connector **410** is moved towards the first guide rail **208**, and away from the straight line A-A'. This will allow the first connector **410** to move over the second connector **424** and pass the second connector **424**. Further, the first suspension **414** is adjusted when the first resilient member **434** resiliently deforms. The first resilient member **434** is arranged to resiliently deform when the outer wall **602** of the first connector **410** pushes against the outer wall **604** of the second connector **424**.

Referring to FIGS. **7**A and **7**B, due to the resilient deformation of the first resilient member **434**, the distance between the first connector **410** and the first guide rail **208** can be adjusted from the distance D1 to a distance D1'. The distance D1' may correspond to a minimum distance between the first connector **410** and the first guide rail **208** along the z-axis when the first connector **410** is displaced from the normal state. Since the first connector **410** moves towards the first guide rail **208**, the distance D1' is less than the distance D1. The wedge-shaped outer walls **602**, **604** may enable the movement of the first connector **410** towards the first guide rail **208** upon engagement between the first and second connectors **410**, **424**. The resilient deformation of the first resilient member **434** allows the first guide coupler **402** to remain drivingly engaged with the first guide rail **208** even through the first connector **410** is displaced from the normal state. Further, the second connector **424** remains engaged with the feeder **202-2** even though the first connector **410** is displaced. The engagement between the second protrusion **432** and the dock **304** (shown in FIG. **5**) allows the second connector **424** to remain engaged with the feeder **202-2** when the first connector **410** contacts the second connector **424** and passes over the second connector **424**. Therefore, a filament feeding operation to the second connector **424** will not be affected by the passing of the first connector **410** over the second connector **424**. Once the first connector **410** has passed over the second connector **424**, the first resilient member **434** returns to an undeformed state, thereby allowing the first connector **410** to return to the normal state/place. In some cases, the first connector **410** may be engaged with any one of the feeders **202** (e.g., the feeder **202-2**), while the second connector **424** may not be engaged with any of the feeders **202**. The second connector **424** may be required to move past the first connector **410** to engage with any other feeder (e.g., the feeder **202-3**). In such a case, the second suspension **428** of the second selector **218** is adjusted such that the second connector **424** moves towards the second guide rail **210**, and away from the straight line A-A'. This may allow the second connector **424** to move past the first connector **410**. Further, the second suspension **428** is adjusted when the second resilient member **436** resiliently deforms. The second resilient member **436** is arranged to resiliently deform when the outer wall **604** of the second connector **424** pushes against the outer wall **602** of the first connector **410**.

FIG. **8** schematically shows a top view of the filament feeding assembly **200**. As can be seen from FIG. **8**, the first and second selectors **216**, **218** in fact move back and forth along the same trajectory. This means that they will have to pass each other in case the first selector **216** needs to engage with a feeder that is, at a certain moment in time, located at the other side of the second selector **218**. The first and second selectors **216**, **218** are driven by the first and second actuators **224**, **226**, respectively. So, by using just two actuators **224**, **226** and two almost identical selection means (i.e., the first and second selectors **216**, **218**), a very compact and functional filament feeding assembly **200** is realized, which enables a flexible way of feeding six (or any other number) filaments into two Bowden tubes **232**, **234** for feeding the dual extruder print head **102**.

If one of the selectors **216**, **218** encounters an obstruction or the end stop of an axis (A-A and B-B) the counter electro magnetic force detected by the actuators, may be used to indicate the mechanical load on the actuator and may trigger a specific set of controls for controlling the filament feeding assembly **200**.

FIG. **9** schematically shows a front view of part of a filament feeding assembly **900** of the FFF system **100**, according to a further embodiment. The filament feeding assembly **900** is substantially similar to the assembly **200** shown in FIGS. **2A-2C**, and common components are assigned the same reference numerals. Some components (e.g., the drive mechanisms) of the filament feeding assembly **900** are not shown in FIG. **9** for illustrative purposes. The filament feeding assembly **900** comprises a first selector **916** and a second selector **918** equivalent to the first selector **216** and the second selector **218**, respectively. The first selector **916** is movably coupled to the first guide rail **208** and comprises a first connector **910**, a first guide coupler **906** and a first suspension **914** equivalent to the first connector **410**, the first guide coupler **402** and the first suspension **414**, respectively, of the first selector **216**. The second selector **918** is movably coupled to the second guide rail **210** and comprises a second connector **924**, a second guide coupler **920** and a second suspension **928** equivalent to the second connector **424**, the second guide coupler **416** and the second suspension **428**, respectively, of the second selector **218**. The first suspension **914** comprises a first resilient member **934**. The second suspension **928** comprises a second resilient member **936**.

In the filament feeding assembly **900**, each of the first and second resilient members **934**, **936** comprises a coil spring. Specifically, the resilient members **934**, **936** of the first and second selectors **916**, **918** comprise first and second coil springs **938**, **940**, respectively. When the first coil spring **938** selectively compresses, a distance between the first connector **910** and the first guide rail **208** is adjusted. Similarly, when the second coil spring **940** selectively compresses, a distance between the second connector **924** and the second guide rail **210** is adjusted. In some embodiments, the first and second coil springs **938**, **940** are arranged to compress and extend along the adjustment axis D-D', where the adjustment axis D-D' is substantially along the z-axis.

FIG. **10** schematically shows a front view of a filament feeding assembly **1000** of the FFF system **100**, according to another embodiment. The filament feeding assembly **1000** is substantially similar to the assembly **200** shown in FIGS. 2A-2C, and common components are assigned the same reference numerals. Some components (e.g., the drive mechanisms) of the filament feeding assembly **1000** are not shown in FIG. **10** for illustrative purposes. The filament feeding assembly **1000** comprises a first selector **1016** and a second selector **1018** equivalent to the first selector **216** and the second selector **218**, respectively. The first selector **1016** is movably coupled to the first guide rail **208** and comprises a first connector **1010**, a first guide coupler **1006** and a first suspension **1014** equivalent to the first connector **410**, the first guide coupler **402** and the first suspension **414**, respectively, of the first selector **216**. The second selector **1018** is movably coupled to the second guide rail **210** and comprises a second connector **1024**, a second guide coupler **1020** and a second suspension **1028** equivalent to the second connector **424**, the second guide coupler **416** and the second suspension **428**, respectively, of the second selector **218**.

In the filament feeding assembly **1000**, each of the first and second suspensions **1014**, **1028** comprises a telescoping mechanism. Specifically, the first and second suspensions **1014**, **1028** comprises first and second telescoping mechanisms **1038**, **1040**, respectively. When the first telescoping mechanism **1038** selectively compresses, a distance between the first connector **1010** and the first guide rail **208** is adjusted. Similarly, when the second telescoping mechanism **1040** selectively compresses, a distance between the second connector **1024** and the second guide rail **210** is adjusted. In some embodiments, the first and second telescoping mechanisms **1038**, **1040** are arranged to retract and extend along the adjustment axis D-D', where the adjustment axis D-D' is substantially along the z-axis. In some embodiments, the first and second telescoping mechanisms **1038**, **1040** may be actuated through actuators (not shown). The actuators may comprise electrical actuators, pneumatic actuators, hydraulic actuators, or combination thereof. It is noted that in some embodiments only one of the suspensions **1014**, **1028** may comprise a telescoping mechanism while the second one may comprise another type of suspension such as a curved member, as was shown in FIG. **4**, or a coil spring, as was shown in FIG. **9**.

In an embodiment, the filament feeding assembly **200**, **900**, **1000** comprises a controller **1102** (see FIG. **11**) arranged to determine target feeders from the plurality of filament feeders for the connectors. The controller **1102** is arranged to control the actuators to move each of the associated connectors to one of the target feeders.

FIG. **11** schematically shows the controller **1102** of the filament feeding assembly **200** (shown in FIGS. **2A-2C**). The controller **1102** is communicably coupled with the components of the filament feeding assembly **200**. Specifically, the controller is communicably coupled with the plurality of feeders **202**, the first and second selectors **216**, **218** (shown in FIGS. 2A-2C), the first and second actuators **224**, **226** (shown in FIGS. 2A-2C), and the first and second sensors **310**, **312** (shown in FIG. **3**) of each feeder **202**. The controller **1102** is arranged to determine target feeders from the plurality of filament feeders **202** for the connectors **1010**, **1024** (shown in FIG. **10**). The target feeders may be determined by controller **1102** using instructions received by the FFF system **100**, or the controller **1102** may be arranged to communicate with other controllers of the FFF system **100** to receive information relating to identities of target feeders for the selectors **216**, **218** to engage with. The controller **1102** is further arranged to control the actuators **224**, **226** to move each of the associated connectors **1010**, **1024** to one of the target feeders. The controller **1102** is further arranged to determine that the connectors **1010**, **1024** are engaged with the target feeders using, e.g., input received from the sensor **310**, **312**. The controller **1102** may further be arranged to operate the target feeders to feed filament **206** through the connectors **1010**, **1024**.

The controller **1102** may include a processor (not shown) and a memory (not shown) storing executable instructions. The processor may execute the instructions stored in the memory to implement a method or an algorithm.

Referring to FIGS. **10** and **11**, the controller **1102** is arranged to move at least the first connector **1010** towards one of the target feeders. In this embodiment wherein the suspension **1014**, **1028** comprises an active element such as the telescoping mechanisms **1038**, **1040**, the controller **1102** may further be arranged to determine that the first connector **1010** is proximal to the second connector **1024** using, e.g., data received from the two actuators **224**, **226**. The controller **1102** will then control the first suspension **1014**, such that the first connector **1010** is retracted towards the first guide rail **208**, in order to allow the first connector **1010** to pass the second connector **1024**. The controller **1102** will then control the first suspension **1014** such that the first connector **1010** moves to a non-retracted state after the first connector **1010** has passed the second connector **1024**. The non-retracted state of the first connector **1010** may correspond to the normal state where the first connector **1010** is arranged to move along the straight line A-A' in front of the plurality of outlets **204**.

FIG. **12** schematically shows a state diagram **1200** for one of the feeders of the filament feeding assembly **200**. The state diagram may be loaded into the controller **1102** (see FIG. **11**) of the filament feeding assembly **200**, **900**, **1000** for proper operation of the filament feeding assembly **200**, **900**, **1000**. The state diagram **1200** depicts operation of one of the feeders **202** of the filament feeding assembly **200** (shown in FIGS. **2A-2C**). The state diagram **1200** comprises a feeder state diagram **1202** depicting operation of the plurality of feeders **202** and a selector state diagram **1250** depicting operation of the first and second selectors **216**, **218** (shown in FIGS. **2A-2C**). The feeder state diagram **1202** comprises a first feeder state **1204**, where the feeder **202** is not operational. The feeder state diagram **1202** proceeds to a second feeder state **1206**. At the second feeder state **1206**, the feeder **202** has been selected as a target feeder **202** for at least one of the first and second selectors **216**, **218**.

Referring to the selector state diagram **1250**, at a first selector state **1252**, the first and second selectors **216**, **218** may be at any unknown positions. Once, a target feeder is selected for at least one of the first and second selectors **216**, **218** (refer to the second feeder state **1206**), at a second selector state **1254**, the at least one of first and second selectors **216**, **218** is moved towards the respective target feeder **202**. At a third selector state **1256**, the at least one of first and second selectors **216**, **218** is docked with the target feeder **202**. Docking may be confirmed by signals received from the at least one sensor **310**, **312** on the target feeder **202**. Further, there may be a change in the target feeder **202**, in which case, the at least one of first and second selectors **216**, **218** is moved to engage with the changed target feeder **202**.

Referring to the feeder state diagram **1202**, once the at least one of first and second selectors **216**, **218** is docked with the respective target feeder **202**, at a third feeder state **1208**, the filament from the respective target feeder **202** is dispensed. It is noted that the feeder state diagram **1202** may comprise further states following state **1208** related to the loading of filament and the printing steps.

The filament feeding assembly **200**, **900**, **1000** of the present invention may allow selective feeding of any two filaments **206** from a plurality of filaments **206**. The filament feeding assembly **200**, **900**, **1000** comprises the first and second selectors **216**, **218** that are movable in the filament feeding assembly **200**, **900**, **1000**, and are arranged to move to and selectively engage with the plurality of feeders **202** and receive respective filaments **206** therefrom. Each of the first and second selectors **216**, **218** may comprise the respective Bowden tube **232**, **234** that transports the filament **206** received at the respective selector **216**, **218**. The filament **206** may be transported to a secondary feeder and then to the print head **102** of the FFF system **100**, or directly to the print head **102** of the FFF system **100**. In the case the print head **102** of the FFF system **100** is a dual extruder print head, each selector **216**, **218** feeds filaments **206** to one of the extruders. The filament feeding assembly **200**, **900**, **1000** is arranged, such that filament **206** is fed to one of the extruders of the print head **102** through common channels, through one of the selectors **216**, **218**. As a result, characteristics of the filament **206** being fed through to the print head **102**, such as friction values and feed force values, are similar, irrespective of the feeder from where the filament is fed. This may allow the FFF system **100** to extrude the filaments **206** more uniformly. Further, since there is a single path, from a selector, for the filament to traverse, the FFF system **100** comprising the proposed filament feeding assembly **200**, **900**, **1000** may not require a merger. This may further reduce friction that may otherwise occur due to movement of the filament through the merger.

The present invention has been described above with reference to a number of exemplary embodiments as shown in the drawings. Modifications and alternative implementations of some parts or elements are possible and are included in the scope of protection as defined in the appended claims. It should be noted that the above-mentioned embodiments illustrate rather than limit the invention, and that those skilled in the art will be able to design many alternative embodiments. In the claims, any reference signs placed between parentheses shall not be construed as limiting the claim. Use of the verb "comprise" and its conjugations does not exclude the presence of elements or steps other than those stated in a claim. The article "a" or "an" preceding an element does not exclude the presence of a plurality of such elements. The mere fact that certain measures are recited in mutually different dependent claims does not indicate that a combination of these measures cannot be used to advantage.

The invention claimed is:

1. A filament feeding assembly for a fused filament fabrication system the filament feeding assembly comprising

a plurality of filament feeders arranged next to each other in a row, wherein each filament feeder comprises an outlet and wherein the respective outlets of the filament feeders are aligned on a straight line;

a first guide rail and a second guide rail, both being arranged parallel to the straight line;

a first selector comprising a first connector movably coupled to the first guide rail and comprising a first filament entrance for selectively receiving a filament from one of the outlets, and a first suspension arranged to adjustably couple the first connector to the first guide rail, so that a distance between the first guide rail and the first connector is adjustable;

a second selector comprising a second connector movably coupled to the second guide rail and comprising a second filament entrance for selectively receiving a filament from one of the outlets, and a second suspension arranged to adjustably couple the second connector to the second guide rail, so that a distance between the second guide rail and the second connector is adjustable; and two actuators, each being arranged to move one of the connectors in front of the respective outlets of the filament feeders;

wherein the first and second connectors are arranged to pass each other by way of adjusting the distance between the first connector and the first guide rail and/or adjusting the distance between the second connector and the second guide rail.

2. The filament feeding assembly according to claim **1**, wherein each of the first and second connectors comprises a wedge-shaped outer wall, and wherein each of the first and second suspensions comprises a resilient member, and wherein the resilient member is arranged to resiliently deform when the outer wall of the first connector pushes against the outer wall of the second connector.

3. The filament feeding assembly according to claim **2**, wherein the resilient member comprises a curved member fixedly coupled to the respective connector.

4. The filament feeding assembly according to claim **2**, wherein the resilient member comprises a coil spring.

5. The filament feeding assembly according to claim **1**, wherein at least one of the first and second suspensions comprises a telescoping mechanism.

6. The filament feeding assembly according to claim **1**, wherein each of the filament feeders comprises a dock, and wherein each of the connectors comprises a protrusion arranged to be received in the dock of the filament feeder.

7. The filament feeding assembly according to claim **1**, wherein each of the filament feeders comprises a sensor arranged to generate a signal indicative of an engagement of one of connectors with the filament feeder.

8. The filament feeding assembly according to claim **1**, further comprising a controller arranged to:

determine target filament feeders from the plurality of filament feeders for the connectors;

control the actuators to move each of the associated connectors to one of the target feeders;

determine that the connectors are engaged with the target feeders; and

operate the target feeders to feed filament through the connectors.

9. The filament feeding assembly according to claim **8**, wherein the controller is further arranged to:

move at least the first connector towards one of the target feeders;

determine that the first connector is proximal to the second connector;

control the first suspension, such that the first connector is retracted towards the first guide rail, in order to allow the first connector to pass the second connector; and

control the first suspension, such that the first connector moves to a non-retracted state after the first connector has passed the second connector.

10. A fused filament fabrication system comprising the filament feeding assembly according to claim **1**.

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